
Solid waste
management

WASTE

- It is defined as

  Waste (also known as rubbish, trash, refuse, garbage, junk) is any unwanted or useless materials.

  OR

  Any materials unused and rejected as worthless or unwanted and “A useless or profile less activity using or expanding or consuming thoughtlessly or carefully.”

Introduction:

Waste:

Depending on their physical state they are classified as:

- Liquid wastes
- Gaseous wastes
- Solid wastes.

Solid waste: Solid waste is the unwanted or useless solid materials generated from combined residential, industrial and commercial activities in a given area.

[Ref: T. Srinivas, Environmental Biotechnology, www.cyen.org]

Solid Waste Management:

Solid waste management is the process of collection, transportation and disposal of solid waste in a systematic, economic and hygienic manner.

'Or'

Solid-waste management is the process of the collecting, treating, and disposing of solid material that is discarded because it is of no longer use.

[Ref: T. Srinivas, Environmental Biotechnology, www.britannica.org]

- In a nutshell “ Solid waste can be regarded as refuse or waste from any kind of source”. But any refuse or waste can be economic resource to others.

Kinds of Wastes

Solid wastes: wastes in solid forms, domestic, commercial and industrial wastes Examples: *plastics , bottles, cans, papers, scrap iron, and other trash*

Liquid Wastes: wastes in liquid form Examples: *domestic washings, chemicals, oils, waste water from ponds, manufacturing industries and other sources.*

Bio-degradable :can be degraded (paper, wood, fruits and others)

Non-biodegradable :cannot be degraded (plastics, bottles, old machines, cans, Styrofoam containers and others)

Hazardous wastes: Substances unsafe to use commercially, industrially, agriculturally, or economically and have any of the following properties- ignitability, corrosivity, reactivity & toxicity.

Non-hazardous : Substances safe to use commercially, industrially, agriculturally, or economically and do not have any of those properties mentioned above. These substances usually create disposal problems.



Classification of wastes according to their origin and type

- **Municipal Solid wastes:** Solid wastes that include household garbage, rubbish, construction & packaging materials, trade refuges etc. are managed by any municipality.
- **Bio-medical wastes:** Solid or liquid wastes including containers, products generated during diagnosis, treatment & research activities of medical sciences.
- **Industrial wastes:** Liquid and solid wastes that are generated by manufacturing & processing units of various industries like chemical, petroleum, coal, metal gas, sanitary & paper etc.
- **Agricultural wastes:** Wastes generated from farming activities. These substances are mostly biodegradable.
- **Fishery wastes:** Wastes generated due to fishery activities.
- **E-wastes:** Electronic wastes generated from any modern establishments. They may be described as discarded electrical or electronic devices. Some electronic scrap components, such as CRTs, wires, circuits, mobile, computers etc.



Basic terms related to solid waste

1. **Ash :** the non-combustible solid by-products of incineration or other burning process.
2. **Bulky waste:** large wastes such as appliances, furniture, and trees and branches, that cannot be handled by normal MSW processing methods.
3. **Co-disposal:** the disposal of different types of waste in one area of a landfill or dump. For instance, sewage sludges may be disposed of with regular solid wastes.

4. **Biodegradable material :** any organic material that can be broken down by microorganisms into simpler, more stable com-pounds. Most organic wastes (e.g., food, paper) are biodegradable.
5. **Compost :** the material resulting from com posting. Compost, also called humus, is a soil conditioner and in some instances is used as a fertilizer.
6. **Composting :** biological decomposition of solid organic materials by bacteria, fungi, and other organisms into a soil-like product.

7. **Disposal :** the final handling of solid waste, following collection, processing, or incineration. Disposal most often means placement of wastes in a dump or a landfill.
8. **Environmental impact assessment (EIA) :** an evaluation designed to identify and predict the impact of an action or a project on the environment and human health and well-being. Can include risk assessment as a component, along with economic and land use assessment.

9. Environmental risk assessment (EnRA) : an evaluation of the interactions of agents, humans, and ecological resources. Comprised of human health risk assessment and ecological risk assessment, typically evaluating the probabilities and magnitudes of harm that could come from environmental contaminants.

10.**Garbage :** in everyday usage, refuse in general. Some MSWM manuals use garbage to mean "food wastes," although this usage is not common.

11.**Landfilling :** the final disposal of solid waste by placing it in a controlled fashion in a place intended to be permanent. The Source Book uses this term for both controlled dumps and sanitary landfills.

12.Leachate : liquid that has seeped through a landfill or a compost pile and has accumulated bacteria and other possibly harmful dissolved or suspended materials.

13. MSW : municipal solid waste.

14. MSWM : municipal solid waste management.

15.Putrescible : subject to decomposition or decay. Usually used in reference to food wastes and other organic wastes that decay quickly.

16.Refuse : all kinds of wastes in solid state excepting excreta from residential, commercial and industrial area.

17.Refuse-derived fuel (RDF) : fuel produced from MSW that has undergone processing. Processing can include separation of recyclables and non-combustible materials, shredding, size reduction, and pelletizing.

18.Rubbish : a general term for solid waste. Sometimes used to exclude food wastes and ashes.

19. **Waste-to-energy (WTE) plant :** a facility that uses solid waste materials (processed or raw) to produce energy. WTE plants include incinerators that produce steam for district heating or industrial use, or that generate electricity; they also include facilities that convert landfill gas to electricity.

Kinds of Wastes

- **Solid wastes:** domestic, commercial and industrial wastes especially common as co-disposal of wastes

  Examples: *plastics, containers, bottles, cans, papers, scrap iron, and other trash*

- **Liquid Wastes:** wastes in liquid form

  Examples: *domestic washings, chemicals, oils, waste water from ponds, manufacturing industries and other sources*

Classification of Wastes according to their Properties

- **Bio-degradable**

  can be degraded (paper, wood, fruits and others)

- **Non-biodegradable**

  cannot be degraded (plastics, bottles, old machines, cans, containers and others)

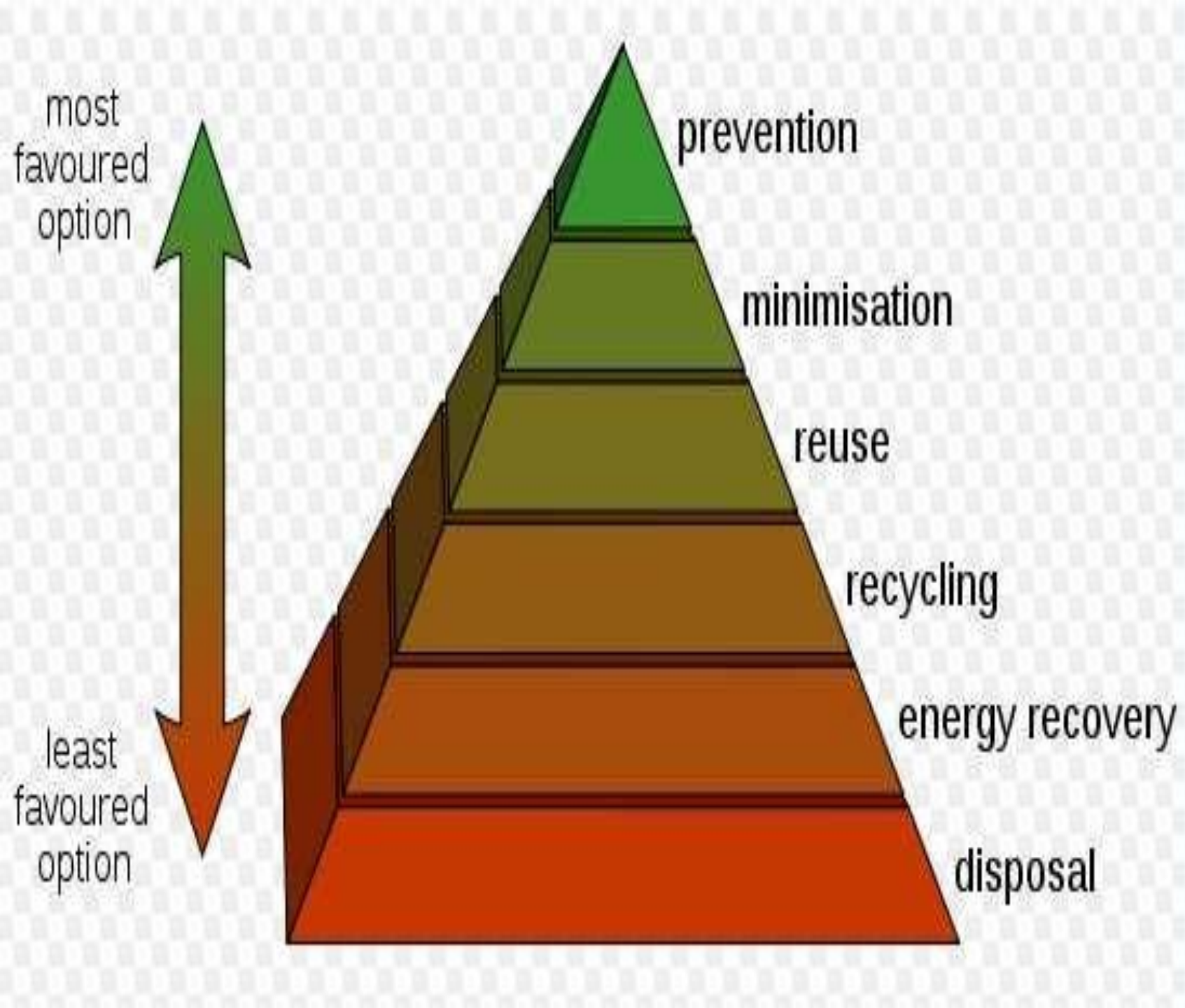
Classification of Wastes according to their Effects on Human Health and the Environment

- **Hazardous wastes**

waste that is reactive, toxic, corrosive, or otherwise dangerous to living things and/or the environment. Many industrial by-products are hazardous.

- **Non-hazardous**

Substances safe to use commercially, industrially, agriculturally, or economically.


most favoured option
least favoured option
prevention
minimisation
reuse
recycling
energy recovery
disposal

CONTD…

- Polluter pays principle - the Polluter Pays Principle is a principle where the polluting party pays for the impact caused to the environment. With respect to waste management, this generally refers to the requirement for a waste generator to pay for appropriate disposal of the unrecoverable material.

SOLID WASTE MANAGEMENT

Waste management is the

- storage
- collection
- transport and handling
- recycling
- disposal and monitoring of waste materials.

STORAGE

- Storage:
  - Galvanized steel dust bin
  - Paper sack
  - Public bins





COLLECTION

- Collection
  - House-to-house collection
  - Collection from the public bins



WASTE HANDLING AND TRANSPORT

- Waste handling and separation involves activities associated with waste management until the waste is placed in storage containers for collection. Handling also encompasses the movement of loaded containers to the point of collection.
- waste is transferred from a smaller collection vehicle to larger transport equipment



RECYCLING

- Recycling refers to the collection and refuse of waste materials such as empty beverage container.
- The materials from which the items are made can be processed into new products.
- Materials for recycling may be collected separately from general waste using dedicated bins.

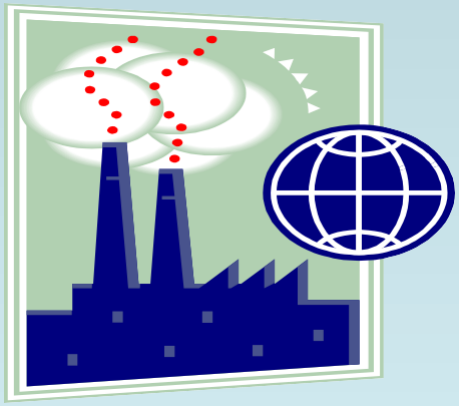


Sources of Wastes





Households



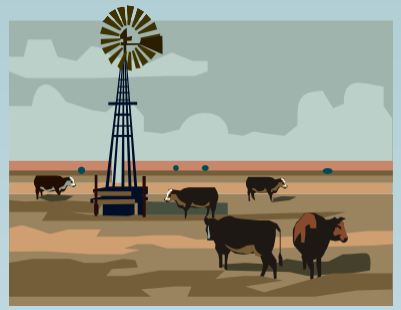


Industry

Sources of Wastes

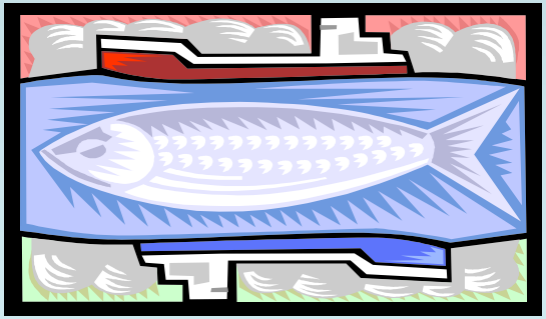
Agriculture



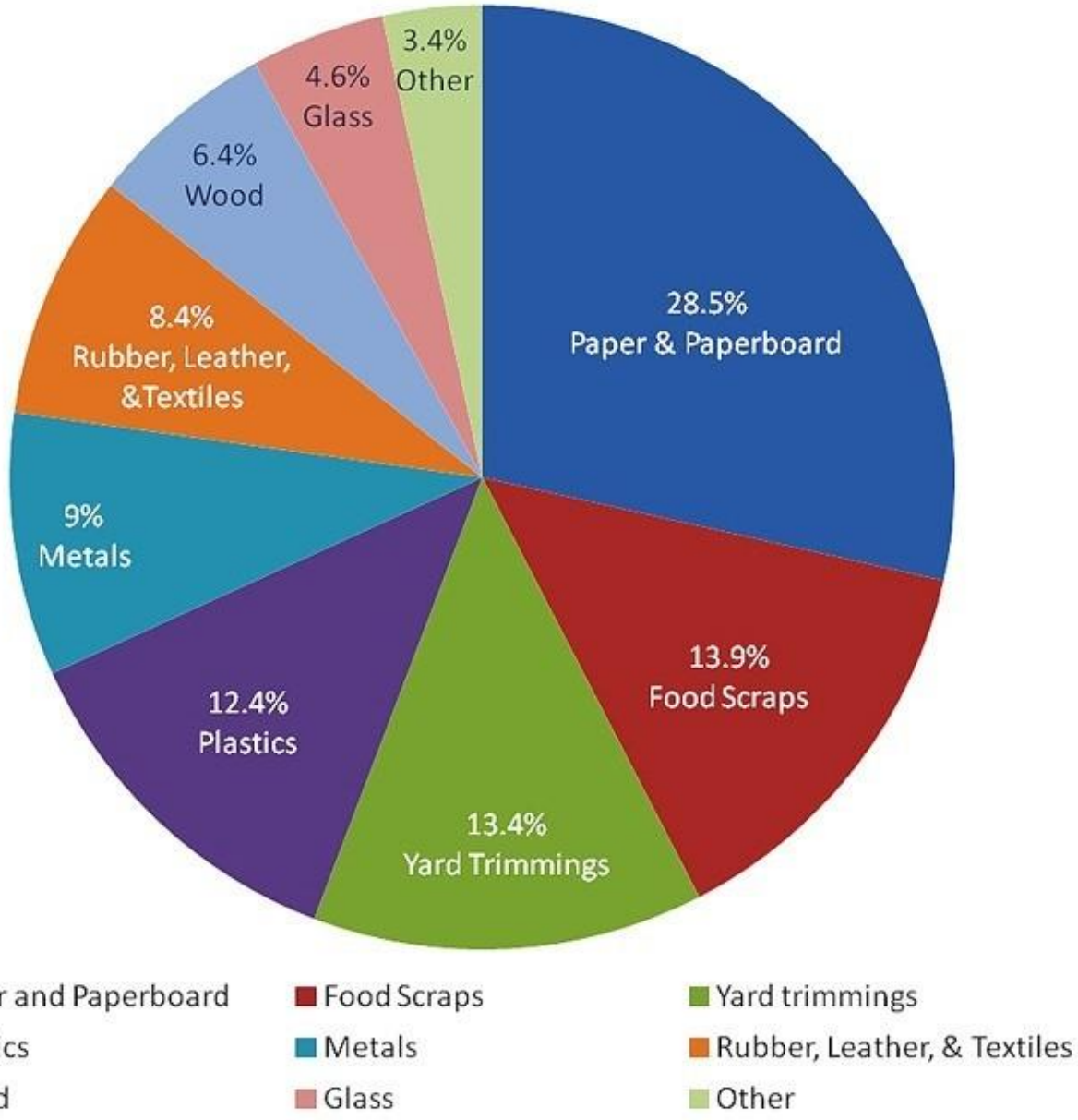


Fisheries





Sources of Wastes



Sources and Types of Solid Wastes

Table 1: Sources and Types of Solid Wastes within a Community

Source	Typical facilities, activities, locations where wastes are generated	Types of solid wastes
Residential	Single and multifamily dwellings	Food wastes, paper, cardboard, plastics, textiles, leather, yard wastes, wood, glass, metals, ashes, special wastes (e.g., bulky items, consumer electronics, white goods, batteries, oil, tires), and household hazardous wastes
Industrial	Light and heavy manufacturing, fabrication, construction sites, power and chemical plants	Industrial process waste, scrap materials, etc. Non - industrial waste including food wastes, construction and demolition wastes, rubbish, ashes , hazardous wastes, ashes, special wastes

Sources and Types of Solid Wastes

Table 1: Sources and Types of Solid Wastes within a Community (Cont'd)

Source	**Typical facilities, activities, locations where wastes are generated**	**Types of solid wastes**
Commercial	Stores, hotels, restaurants, markets, office buildings, etc.	Paper, cardboard, plastics, wood, food wastes, glass, metals, special wastes, hazardous wastes
Institutional	Schools, hospitals, prisons, government centers	Same as commercial
Construction and Demolition	New construction sites, road repair, renovation sites, demolition of buildings, broken pavement	Wood, steel, concrete, dirt, etc.

Sources and Types of Solid Wastes

Table 1: Sources and Types of Solid Wastes within a Community (Cont'd)

Source	Typical facilities, activities, locations where wastes are generated	Types of solid wastes
Municipal Services (excluding treatment facilities)	Street cleaning, landscaping, parks, beaches, other recreational areas, water and wastewater treatment plants	Street sweepings; landscape and tree trimmings; general wastes from parks, beaches, and other recreational areas; sludge
Agricultural	Field and row crops, orchards, vineyards, dairies, feedlots, farms, etc.	Spoiled food wastes, agricultural wastes, rubbish, hazardous waste.

Solid Waste in India

- 7.2 million tonnes of hazardous waste
- One Sq km of additional landfill area every-year
- Rs 1600 crore for treatment & disposal of these wastes
- In addition to this industries discharge about 150 million tonnes of high volume low hazard waste every year, which is mostly dumped on open low lying land areas.

Source: Ministry of Environment & Forest

Growth of Solid Waste In India

- In 1981-91, population of Mumbai increased from 8.2 million to 12.3 million
- During the same period, municipal solid waste has grown from 3200 tonnes to 5355 tonnes, an increase of 67%
- Waste collection is very low for all Indian cities
- City like Bangalore produces 2000 tonnes of waste per annum, the ever increasing waste has put pressure on hygienic condition of the city

- *Source: The Energy & Resources Institute, New Delhi*

- Estimated waste generation is 1,00,000 MT/day.
- Per capita waste generation ranges between 0.20 to 0.60 kg.
- Waste collection efficiency in bigger sized cities ranges from 70 to 90% and in small sized towns it is up to 50-60%.
- Local authorities spend less 5% of their budget on waste disposal and maximum cost is incurred on street sweeping and collection and transportation of waste.

MAJOR DEFICIENCIES

- Littering of garbage due to unorganized primary collection
- Provision and operation of interim storage facilities unsatisfactory
- Irregular garbage lifting
- Transportation system not synchronize with storage facilities
- Processing/ treatment of MSW not practiced
- Final disposal through dumping and not SLF

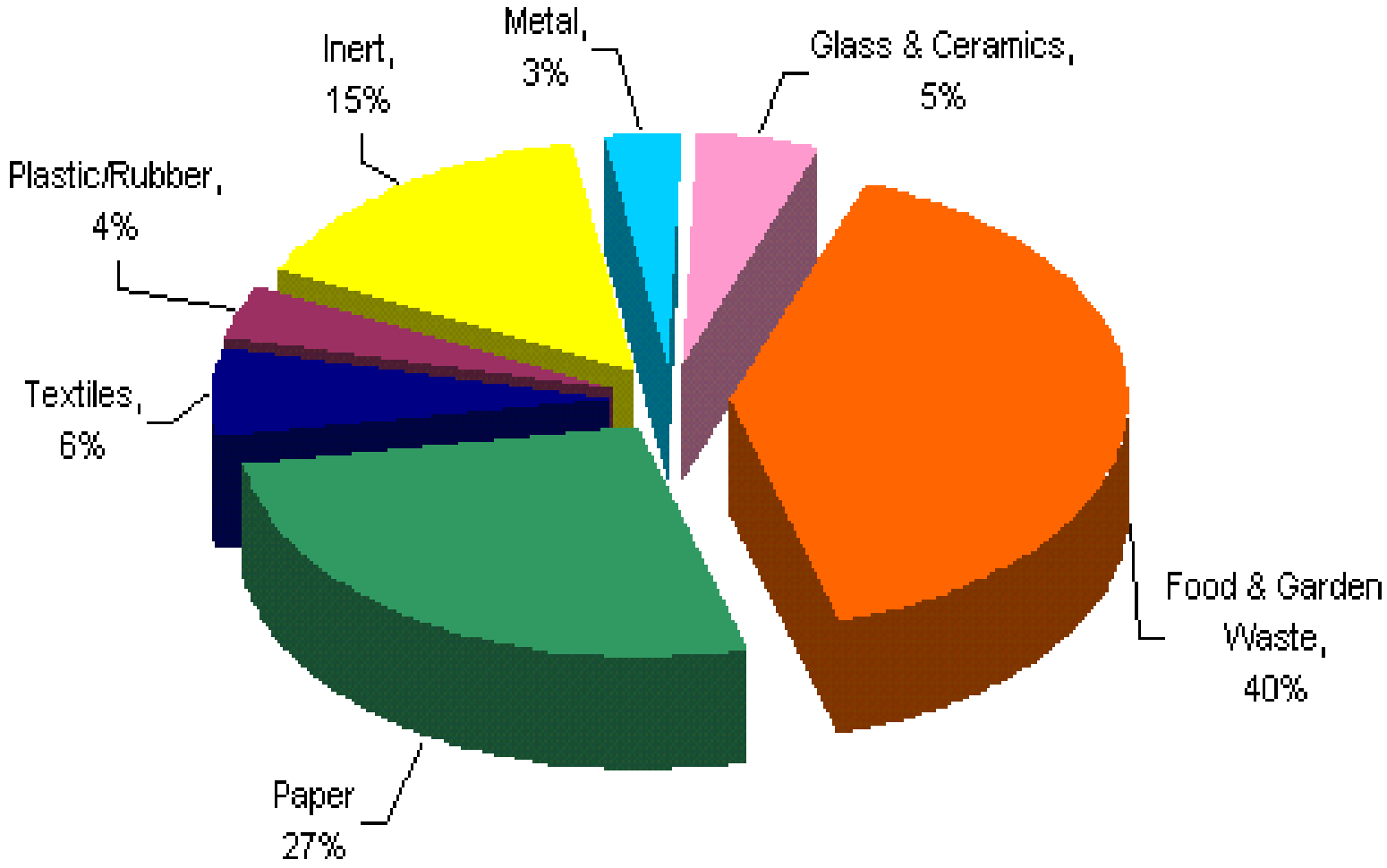
Effects of waste if not managed wisely

- Affects our health
- Affects our socio-economic conditions
- Affects our coastal and marine environment
- Affects our climate

Composition of solid waste

- The general composition of solid waste being generated from the cities of India is 40% Food & Garden waste, 5% glass & Ceramics, 3% Metal, 15% inert, 4% Plastic/ Rubber, 6 % Textile, 27 % Paper.

- Total Organic Fraction - 40%, Combustible Fraction - 37%, Recyclables - 8%, Inert - 15%



Source: CPHEEO Manual on MSW, 2005

Characteristics of solid waste

Three types of characteristics:

1. Physical
2. Chemical and
3. Biological

Physical characteristics

- This includes the determination of percent contents of various ingredients of the solid waste.
- Bulk Density is generally calculated.
- Function of location, season, storage time, equipment used, processing (compaction, shredding, etc.)
- Used in volume calculations.

Chemical characteristics

- Used primarily for combustion and waste to energy (WTE) calculations but can also be used to estimate biological and chemical behaviours.

- Waste consists of combustible (i.e. paper) and non-combustible materials (i.e. glass).

Proximate Analysis

- Loss of moisture (temp held at 105 C)
- Volatile Combustible Matter (VCM) (temp increased to 950 C, closed crucible)
- Fixed Carbon (residue from VCM)
- Ash (temp = 950C, open crucible)

Fusing Point of Ash

- Clinker (agglomerations of carbon and metals) formation temperature, 2000 to 2200 F

Ultimate Analysis

- Molecular composition (C, H, N, O, P, etc.)

Energy Content

- Determined through lab calculations using calorimeters

Biological characteristics

Biodegradability

- Organic fraction often equated with the volatile solids (VS) content of the waste
  - However, not all organic materials are easily degradable
- Biodegradable fraction -
- Degradation produces odours
  - Hydrogen sulfide, H_2S (rotten eggs)
  - Methyl mercaptans
  - Aminobutyric acid
  - Methane is odourless.
- Attracts flies, vermin, rodents (vectors)

Solid waste collection and transport

Factors considered:

i) Types of Containers:

- Depend on:
  - characteristics of SW collected
    - E.g. Large storage containers (Domestic SW: flats/apartment)
    - Containers at curbs
    - Large containers on a roller (Commercial/Industrial)
  - Collection frequency
  - Space available for the placement of containers

- Residential; refuse bags (7 -10 litres)
- Rubbish bins - 20 -30 litres
- Large mechanical containers - more commonly used to cut costs (reduce labor, time , & collection costs)
- must be standardized to suit collection equipment.

ii) Container Locations:

- side/rear of house
- alleys
- special enclosures (apartment/condos)
- Basement (apts. in foreign countries)/ newer complexes

iii) Public Health:

- relates to on-time collection to avoid the spread of diseases by vectors, etc.

iv) Aesthetics:

- must be pleasing to the eye (containers must be clean, shielded from public's view).

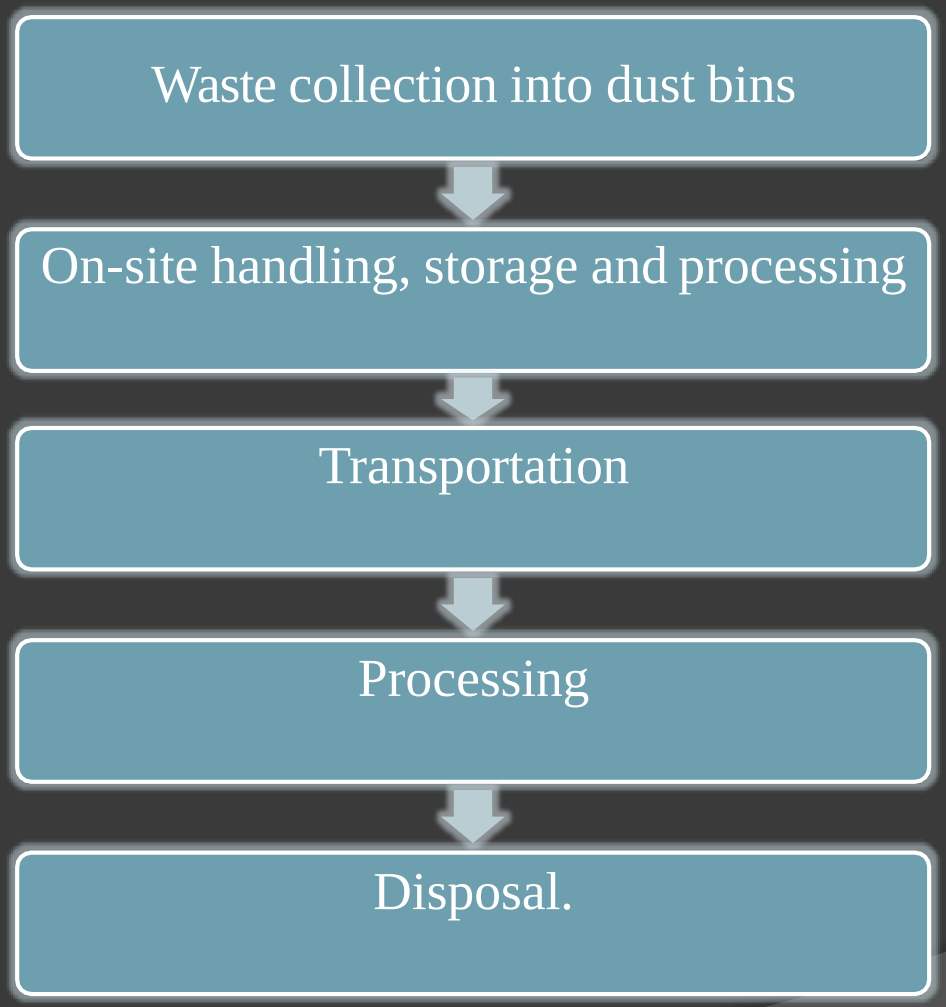
v) Collection of SW

- 60-80 percent of total SWM costs.
- Malaysia (other developing nations) - labor and capital intensive.
- Major problems:
  - Poor building layouts - e.g. squatters
  - Road congestion - time cost, leachate, transport costs.
  - Physical infrastructure
  - Old containers used (leaky/ damaged)
  - Absence of systematic methods (especially at apartments, markets with large wst. volume).

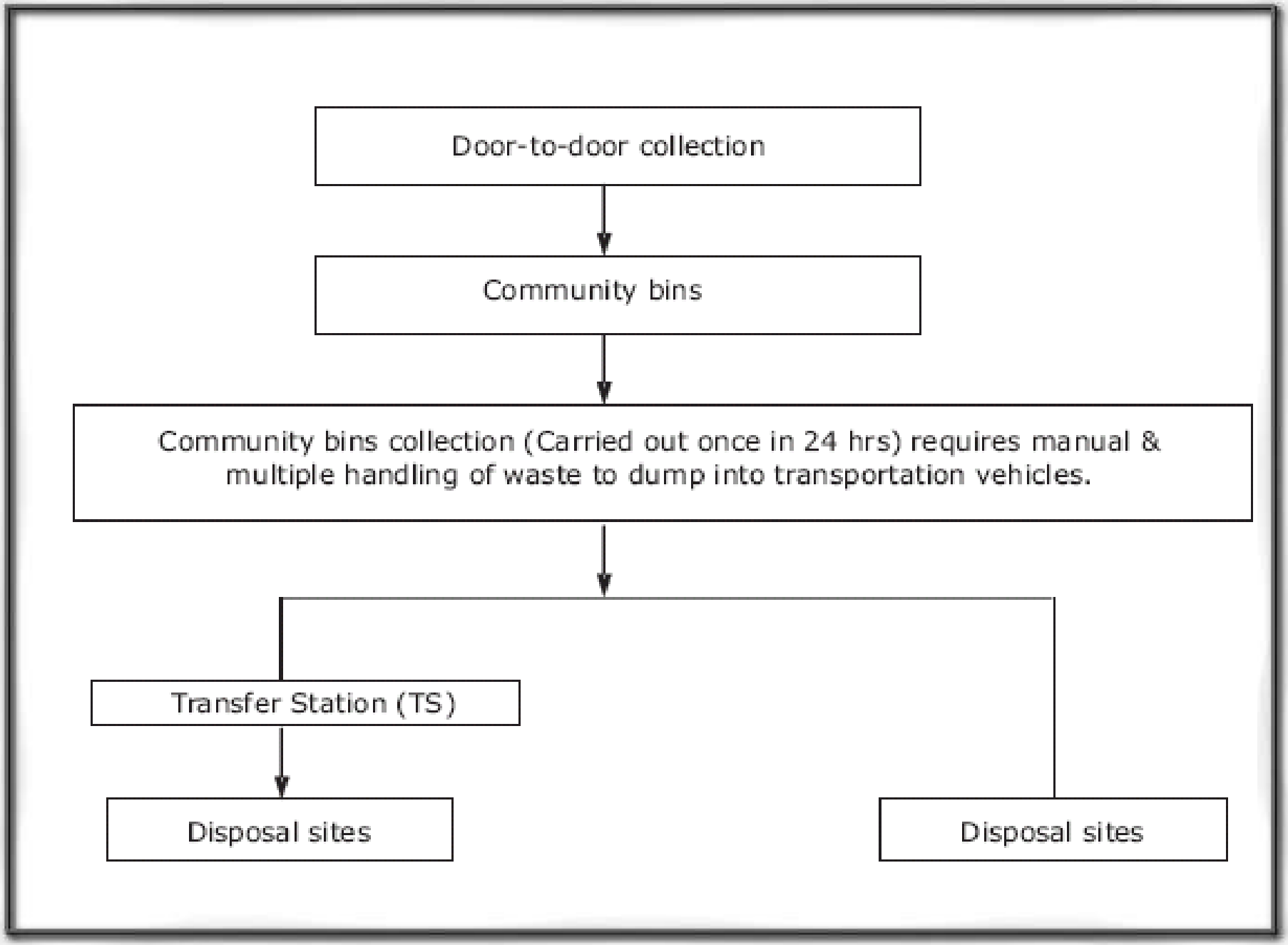
Collections were made by:

1. Municipal/ District Council
2. Private firm under contract to municipal
3. Private firm contract with private residents

3. Engineered systems for solid waste management:



[Ref: T. Srinivas, Environmental Biotechnology]


Door-to-door collection
Community bins
Community bins collection (Carried out once in 24 hrs) requires manual & multiple handling of waste to dump into transportation vehicles.
Transfer Station (TS)
Disposal sites
Disposal sites

Types of collection

Municipal Collection Services:

a. Residential:

1. Curb (*Kerb-side*)

2. Alley

3. Set out and set back

4. Backyard collection

Curb (Kerb-side)

- House owner is responsible for placing solid waste containers at the curb on scheduled day.
- The work man come, collect and empty the container and put back at the curb.
- House owner is required to take back the empty containers from the curb to his house.
- Quickest/ economical
- Crew: 1 driver + 1 or 2 collectors
- No need to enter property

Set-out, set back

- Collectors have to enter property
- Set out crew carries full containers from resident storage location to curb/ alley before collection vehicle arrives.
- Collection crew load their refuse into vehicle
- Set-back crew return the container to storage area.

Alley service

- The containers are placed at the alley line from where they are picked up by workmen from refuse vehicles who deposit back the empty container.

Backyard service

- The workers with the vehicles carry a bin, wheel – barrow or sack or cloth to the yard and empty the solid waste container in it.

- The bin is taken to solid waste vehicles where it is emptied.

Commercial-Industrial Collection Services:

i. Large movable and stationary containers

ii. Large stationary compactors (to form bales)

Collection Frequency:

- residential areas : everyday/ once in 2 days
- communal/ commercial : daily
- food waste - max. period should not exceed :

  - the normal time for the accumulation of waste to fill a container
  - the time for fresh garbage to putrefy and emit fouls odor
  - the length of fly-breeding cycle (< 7 days).


YES
YES
Coquitlam
MIXED PAPER
YES
NO
NEWSPAPER - PUT IN BLUE BAG
Coquitlam
NEWSPRIN
BLUE BOX
YELLOW BAG
BLUE BAG
All material accepted in the Recycling Program is banned from the landfill.

Treatment and disposal of solid waste

Several methods are used for treatment and disposal. These are:

1. Open Dumps
2. Composting
3. Incineration
4. Landfilling
5. Pyrolysis
6. Recycling

i. Open Dumps:

•Open dumps refer to uncovered areas that are used to dump solid waste of all kinds.

• The waste is untreated, and not segregated.

•It is the breeding ground for files, rats, and other insects that spread disease.

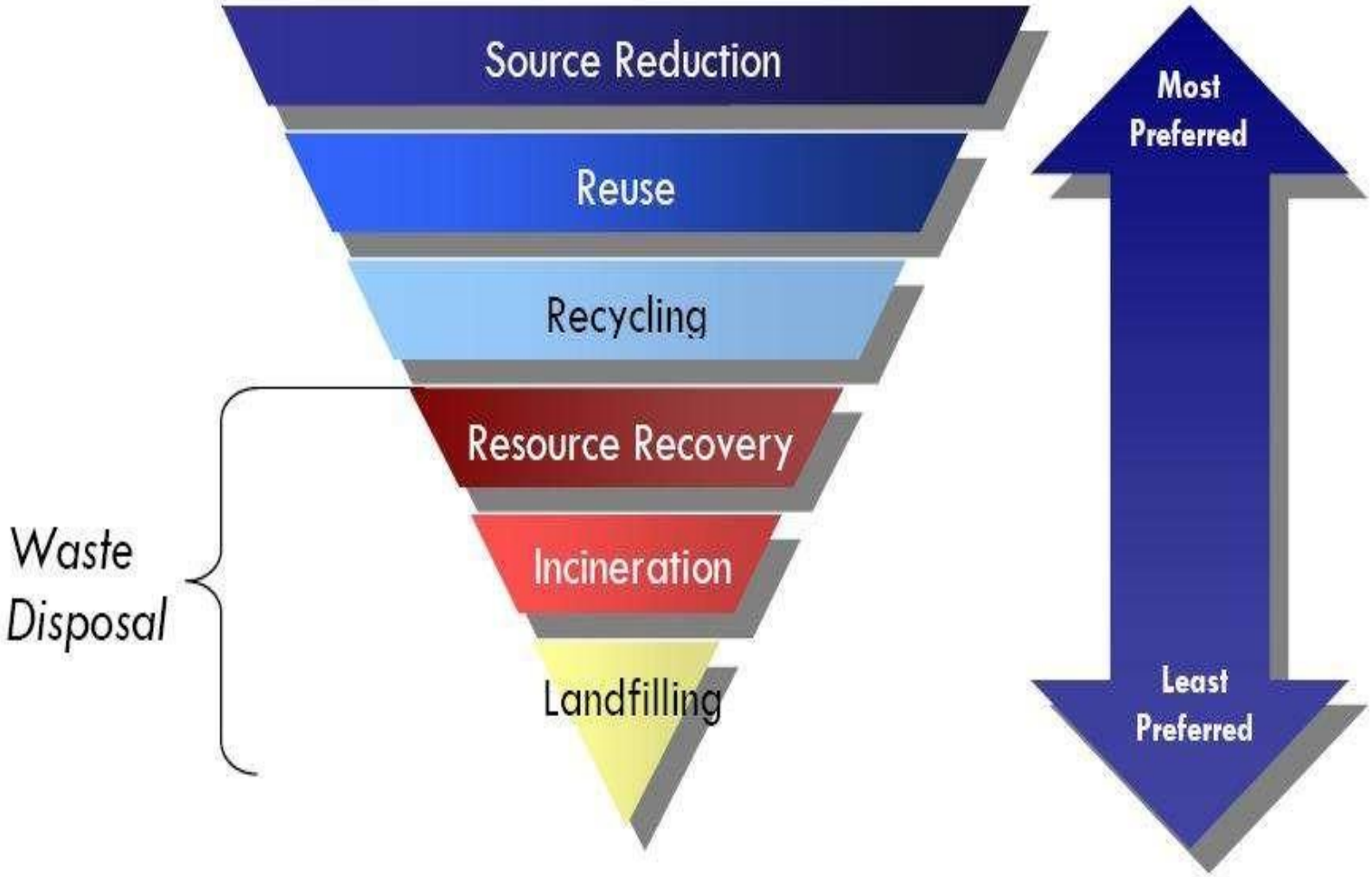
• The rainwater run-off from these dumps contaminates nearby land and water, thereby spreading disease. In some countries, open dumps are being phased out.



Open dump site

[Ref: Indu Shekhar Thakur- Environmental Biotechnology]

Figure 3-1. The Solid Waste Management Hierarchy



iii. Composting:

Composting is the biological decomposition of organic waste under controlled aerobic condition.

Industries as paper, agricultural and food processing give out wastes which are almost 100% organic. This organic matter can be composted to yield good manure.

Compost is the end product obtained after subjecting the organic fraction of solid waste to aerobic or anaerobic decomposition to yield humus like solid, carbon dioxide, water vapour and energy.

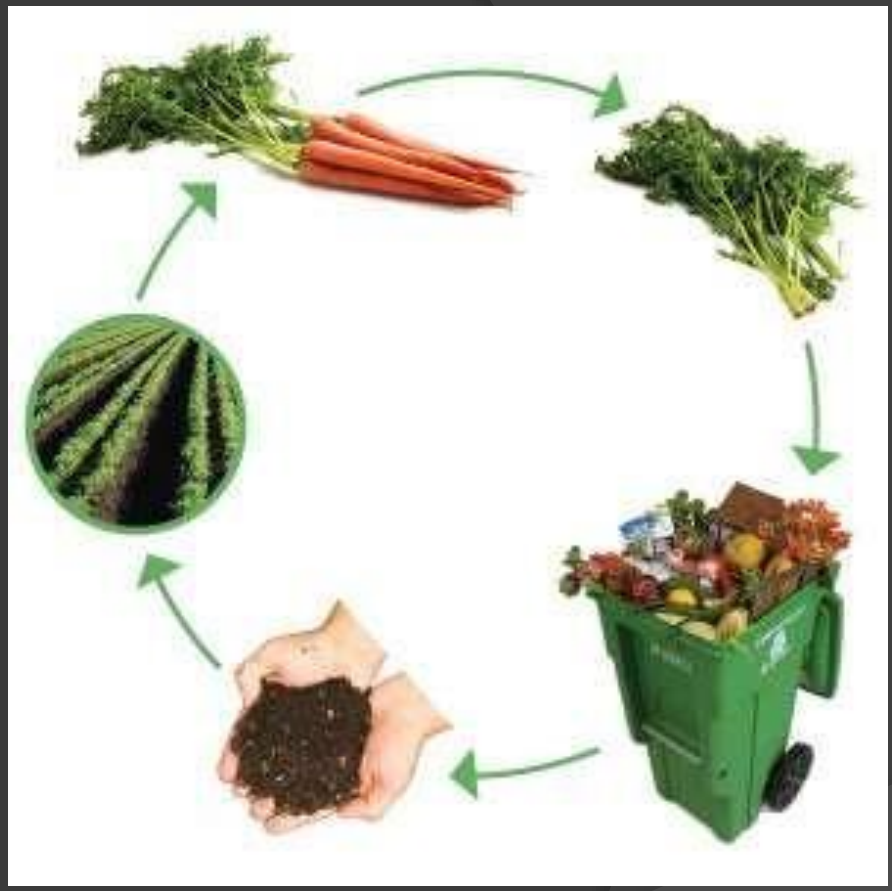


Fig: Compost cycle

[Ref: Indu Shekhar Thakur- Environmental Biotechnology and www.cuesa.org]

➢ Different stages of composting:

- ❑ Segregation of solid waste
- ❑ Processing the compostable matter
- ❑ Preparation for compost
- ❑ Digestion
- ❑ Curing
- ❑ Screening

$$\text{Organic matter} + O_2 \xrightarrow{\text{Aerobic bacteria}} CO_2 + H_2O + NH_3 + \text{energy} + \text{other end products}$$

$$\text{Organic matter} \xrightarrow{\text{Anaerobic bacteria}} CH_4 + CO_2 + H_2O + NH_3 + H_2S + \text{energy} + \text{other end products}$$

[Ref: T. Srinivas, Environmental Biotechnology]

➢ Mechanism of composting:

• Composting is a very complex process involving the participation of several microorganisms like bacteria, actinomycetes and fungi.

•The bacteria bring out the decomposition of macromolecule namely proteins and lipids. Besides generating energy (heat) . Fungi and actinomycetes degrade cellulose and other complex organic compounds.

•Composting may be divided into three stages with reference to changes in temperature**:**

- ❑ Mesophilic stage
- ❑ Thermophilic stage
- ❑ Cooling stage

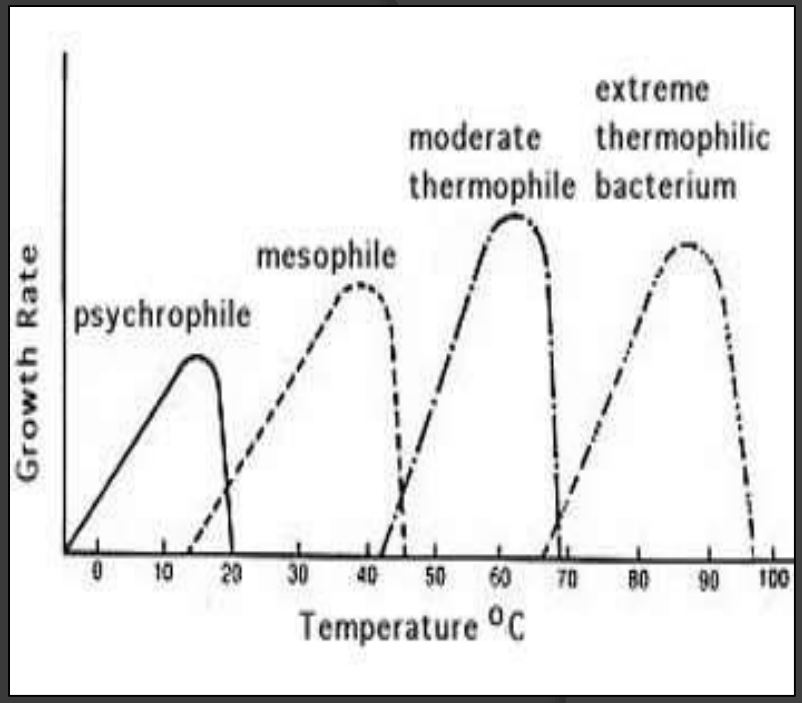


Fig**:** Growth pattern of microbes during composting

[Ref**:** U.Satayanarayan, Biotechnology, and S.V.S Rana, environmental Biotechnology]

A. COMPOSTING

- It is controlled biological decomposition of organic matter, such as food and yard wastes, into humus.
- Composting is the natural process of 'rotting' or decomposition of organic matter by microorganisms under controlled conditions.
- It can be anaerobic and aerobic.
- This process takes about 4 to 6 weeks.

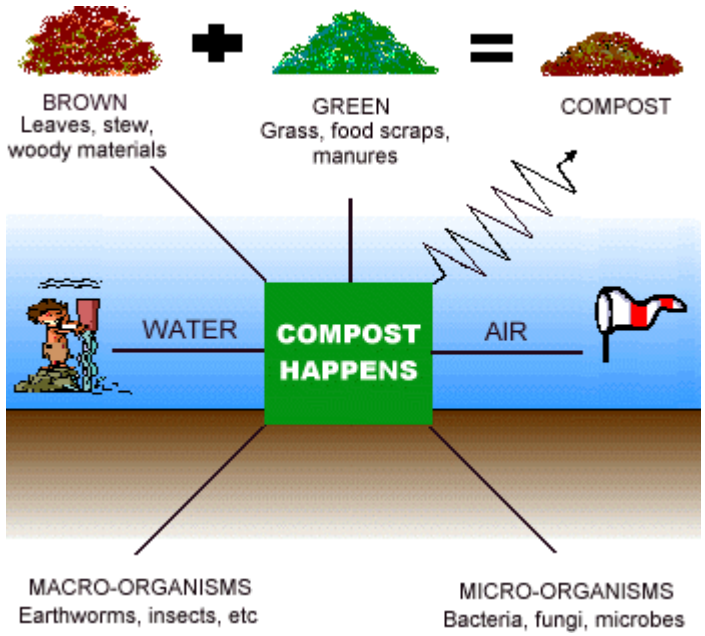
➢ <u>Benefits of composting :</u>

✓ Provides nutrients to the soil.

✓ Increases beneficial soil organisms.

✓ Protects soil from erosion.

✓ Assists pollution remediation.



Composting

- It is a process in which organic matter of solid waste is decomposed and converted to humus and mineral compounds.
- Compost is the end product of composting, which used as fertilizer.
- Three methods of composting:

  (a) composting by trenching

  (b) open windrow composting

  (c) mechanical composting

Composting by trenching

- Trenches 3 - 12 m long, 2 – 3 m wide and 1- 2 m deep with spacing 2 m.
- Dry wastes are filled up in 15 cm. On top of each layer 5 cm thick sandwiching layer of animal dung is sprayed in semi liquid form.
- Biological action starts in 2- 3 days and decomposition starts.
- Solid waste stabilize in 4- 6 months and changed into brown colored odorless powdery form known as humus.





Open windrow composting

- Large materials like broken glass, stone, plastic articles are removed.
- Remaining solid wastes is dumped on ground in form of piles of 0.6 – 1 m height.
- The width and length of piles are kept 1- 2 m and 6 m respectively.
- Moisture content maintained at 60%.
- Temp. increases in side pile.
- After pile for turned for cooling and aeration to avoid anaerobic decomposition.
- The complete process may take 4- 6 week.



Mechanical composting

- It requires small area compare to trenching and open windrow composting.
- The stabilization of waste takes 3- 6 days.
- The operation involved are

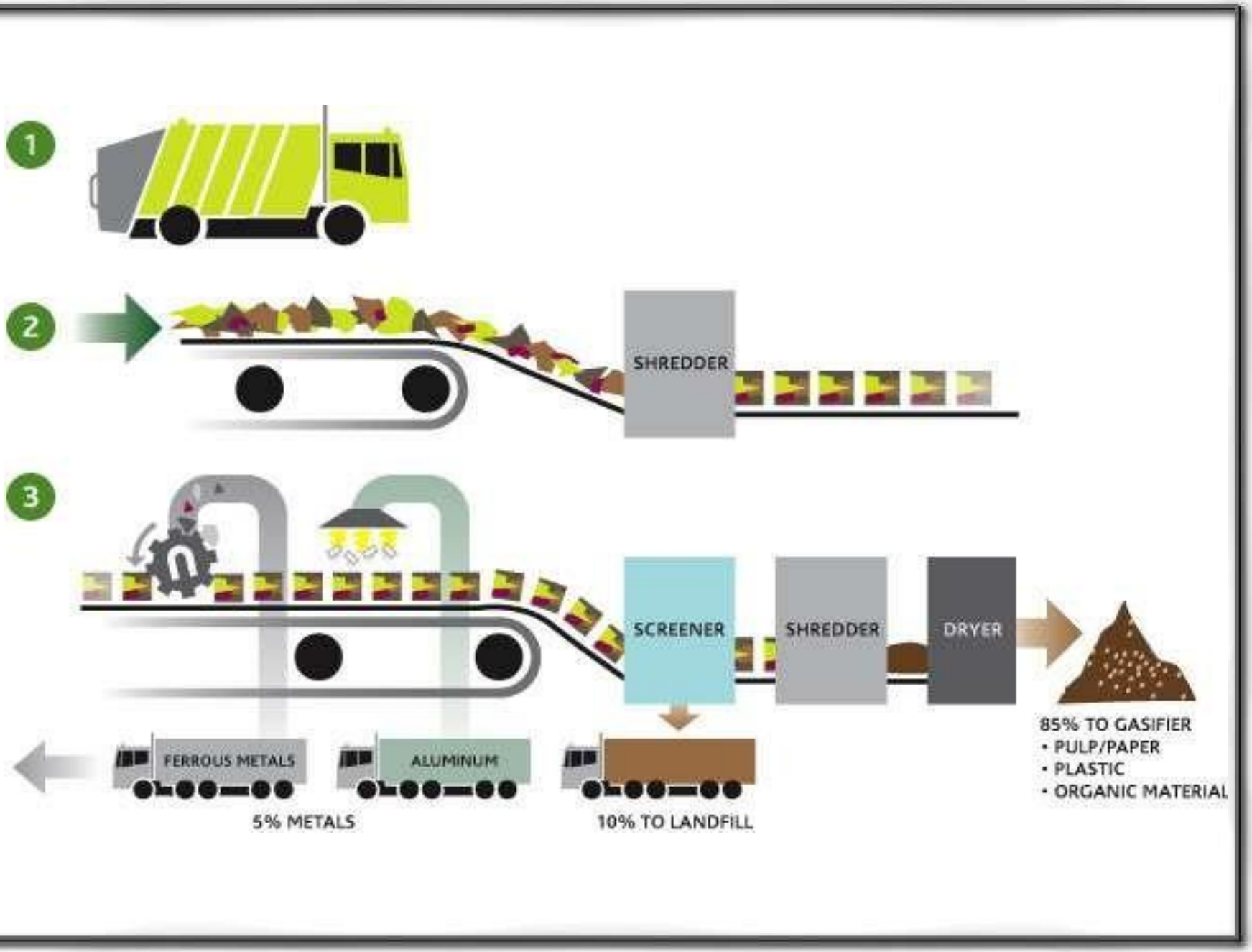
  reception of refuse

  segregation

  shredding

  stabilization

  marketing the humus


1
2
SHREDDER
3
SCREENER
SHREDDER
DRYER
85% TO GASIFIER
• PULP/PAPER
• PLASTIC
• ORGANIC MATERIAL
FERROUS METALS
ALUMINUM
5% METALS
10% TO LANDFILL

A. BANGALORE METHOD (HOT FERMENTATION PROCESS/ ANAEROBIC METHOD)

- IISc- Indian Council of Agricultural Research.
- Trenches are dug 3ft deep, 5-8ft broad, 15-30ft long.
- Composting procedure
  - 1. Layer of Refuse- 15 cm
  - 2. Layer of Night soil – 5 cm

Physical, chemical and biological changes takes place

B. MECHANICAL COMPOSTING (AEROBIC METHOD)

- Compost manufactured by processing raw materials.
- 1. Screening
- 2. Pulverization (<2inches)
- 3. Mixing
- 4. Incubation

Process completed in 4-6 week

C: vermicomposting :It is a method of disposal of kitch*en and* plate wastes, which serves the dual purpose of disposing off the garbage as well as proving eco-friendly.



v. Vermicomposting:

•Vermicomposting is a simple biotechnological process of composting, in which certain species of earthworms are used to enhance the process of waste conversion and produce a better end product.

• Vermicomposting differs from composting in several ways.

- Utilizing microorganism's (earthworms that are active at 10–32°C).
- The process is faster than composting (Because the material passes through the earthworm gut, a significant but not yet fully understood transformation takes place, whereby the resulting earthworm castings (worm manure) are rich in microbial activity and plant growth regulators).

[Ref: Indu Shekhar Thakur- Environmental Biotechnology]

➢ Types of earthworms:

There are nearly 3600 types of earthworms in the world and they are mainly divided into two types**:**

- ❑ **Burrowing earthworm**
- ❑ **Non-burrowing earthworm.**

The Indian Species are: *Dichogaster bolaui, Drawida willsi, Lampito mauritti, perionyx excavates, O.Surensis, M.elongata.*

Characters	Burrowing earthworm	Non-burrowing earthworm
Habitat	Live deep in soil.	Live in upper layer of soil
Colour	Pale	Red, Purple
Length	20-30 cm.	10-15 cm.
Life span	15 years	28 months
Example	*Pertima elongate, Pertima asiatica*	*Eisenia fetida, Eudrilus eugenae*

➢ Vermicomposting process:



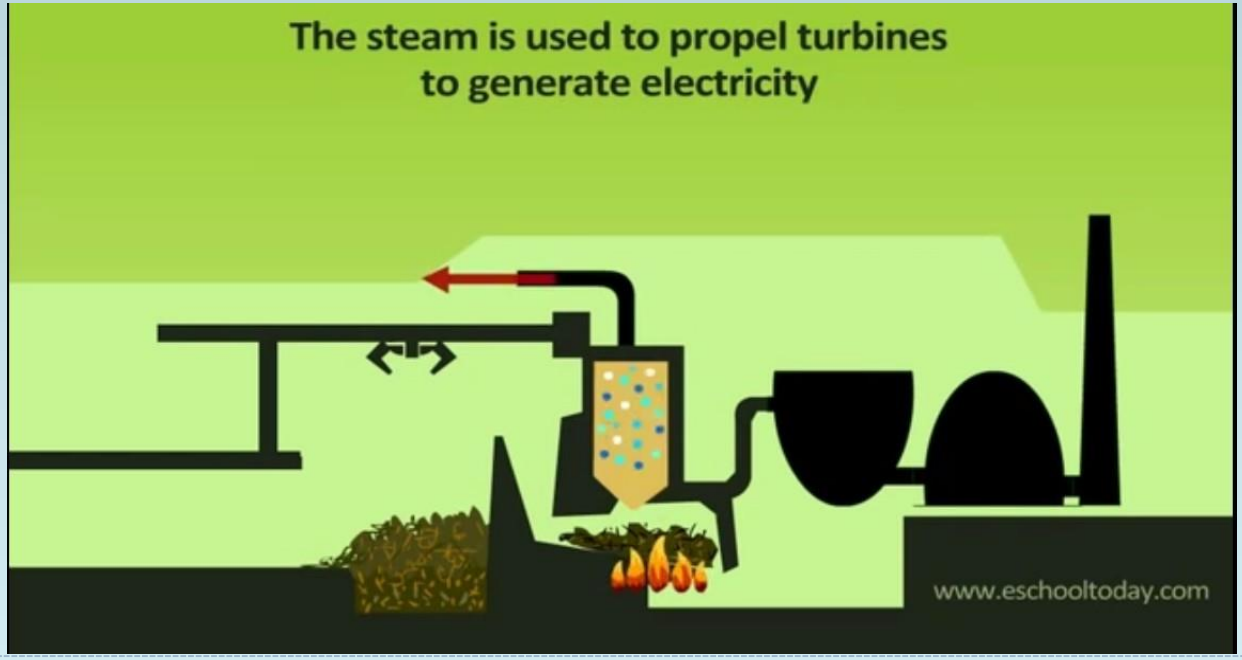


Vermicomposting Process

[Ref: www.icrisat.org]

2. INCINERATION

- Incineration is a waste treatment process that involves the combustion of solid waste at 1000C.
- waste materials are converted into ash, flue gas, and heat.
- The ash is mostly formed by the inorganic constituents of the waste and gases due to organic waste.
- the heat generated by incineration is used to generate electric power.



ADVANTAGES

- Minimum of land is needed compared to other disposal methods.
- The weight of the waste is reduced to 25% of the initial value.
- No risk of polluting local streams and ground waters as in landfills.
- Incineration plants can be located close to residential areas.
- Gases are used to generate power.

DISADVANTAGES

- Expensive
- Required skilled labour.
- The chemicals that would be released into the air could be strong pollutants and may destroy ozone layer (major disadvantage).
- high energy requirement.



C. INCINERATION

- Incineration is a waste management technology that involves the combustion of organic materials and/or substances.
- It is carried out at high temperature.
- The waste material is converted into ash, flue gases, particulates and heat.
- Types of incinerators :

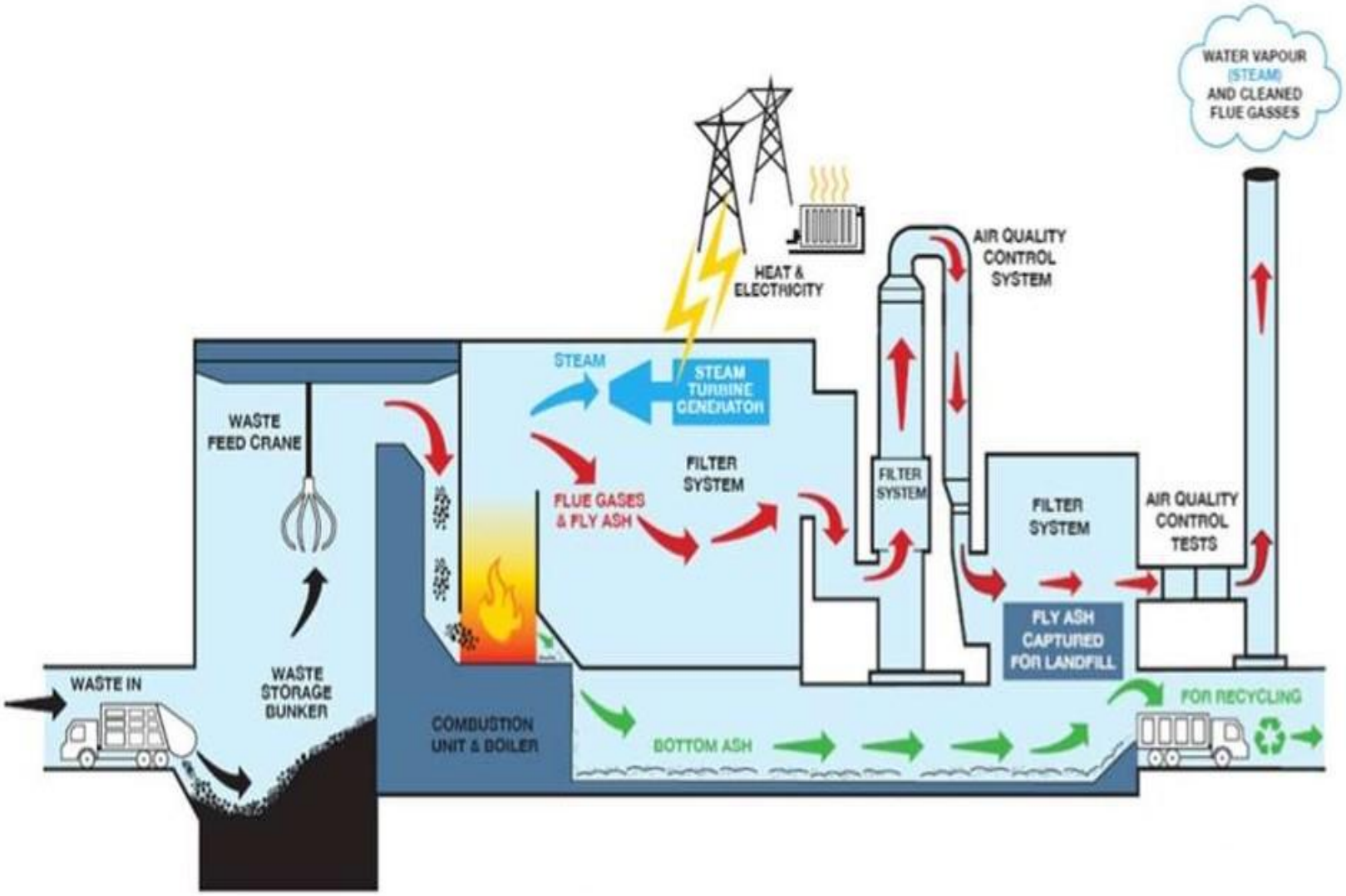
i. Moving grate
ii. Fixed grate
iii. Fluidized bed
iv. Rotary kiln

➢ Advantages :

✓ Less space requirement.
✓ Hygienic process.

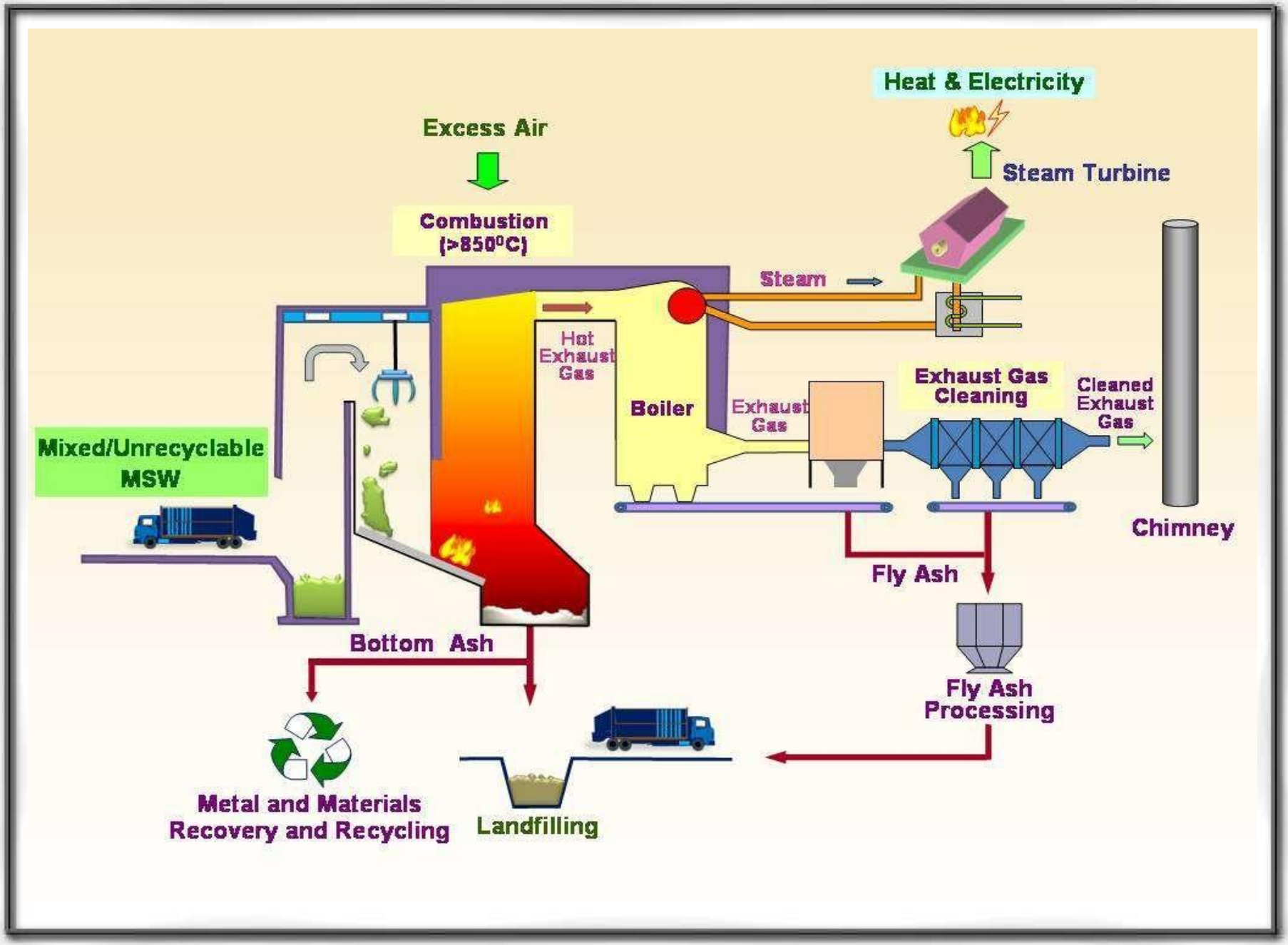
➢ Disadvantages :

✓ Expensive process.
✓ Special care required.


WATER VAPOUR
(STEAM)
AND CLEANED
FLUE GASSES
HEAT &
ELECTRICITY
AIR QUALITY
CONTROL
SYSTEM
STEAM
STEAM
TURBINE
GENERATOR
WASTE
FEED CRANE
FILTER
SYSTEM
FLUE GASES
& FLY ASH
FILTER
SYSTEM
FILTER
SYSTEM
AIR QUALITY
CONTROL
TESTS
FLY ASH
CAPTURED
FOR LANDFILL
WASTE IN
WASTE
STORAGE
BUNKER
COMBUSTION
UNIT & BOILER
BOTTOM ASH
FOR RECYCLING

Incineration

- Incineration is a waste treatment process that involves the combustion of organic substances contained in waste materials.
- Incineration and other high temperature waste treatment systems are described as "thermal treatment".
- Incineration of waste materials converts the waste into ash, flue gas, and heat.
- Incinerators are used for this process.


Heat & Electricity
Excess Air
Steam Turbine
Combustion
(>850°C)
Steam
Hot
Exhaust
Gas
Boiler
Exhaust
Gas
Exhaust Gas
Cleaning
Cleaned
Exhaust
Gas
Mixed/Unrecyclable
MSW
Chimney
Fly Ash
Bottom Ash
Fly Ash
Processing
Metal and Materials
Recovery and Recycling
Landfilling

Important points regarding incineration

- Supplying of solid waste should be continuous.
- Waste should be proper mixed with fuel for complete combustion.
- Temp. should not less than 670 °C.

Advantages

- Most hygienic method.
- Complete destruction of pathogens.
- No odor trouble.
- Heat generated may be used for steam power.
- Clinkers produced may be used for road construction.
- Less space required.
- Adverse weather condition has no effect.

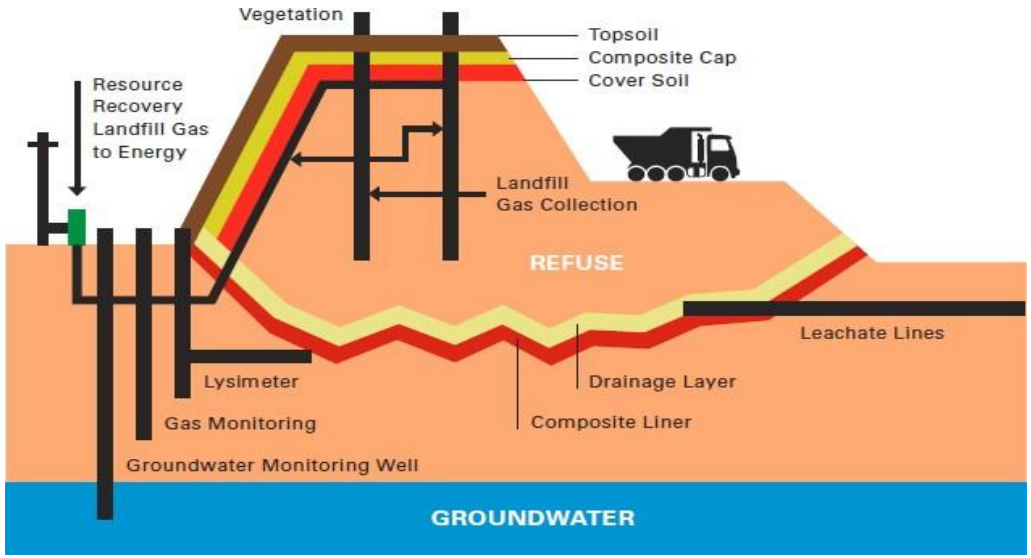
Disadvantages

- Large initial expense.
- Care and attention required otherwise incomplete combustion will increase air pollution.
- Residues required to be disposed which require money.
- Large no of vehicles required for transportation.

B. LANDFILLING

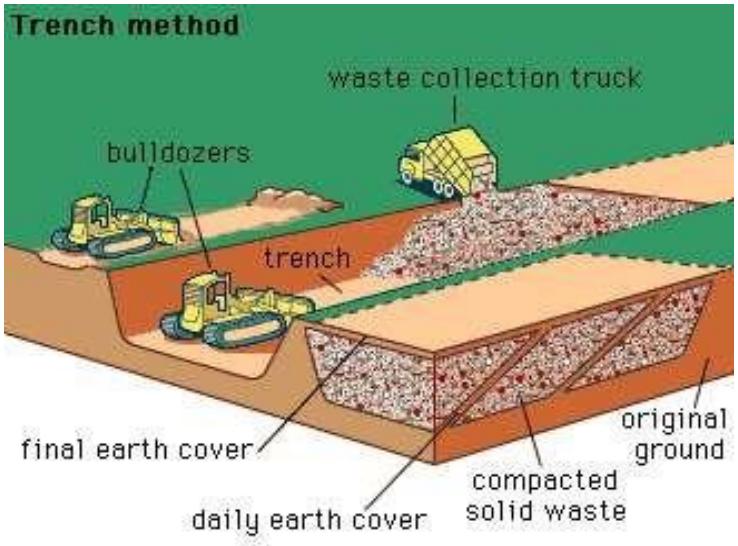
- A landfill site is a site for the disposal of waste materials by burial and is the oldest form of waste treatment.
- Waste is directly dumped into mining voids or borrow pits. Disposed waste is compacted and covered with soil.
- Gases generated by the decomposing waste materials are often burnt to generate power.
- <u>Landfilling can be done by three methods :</u>

1) Trench method
2) Area method
3) Ramp method



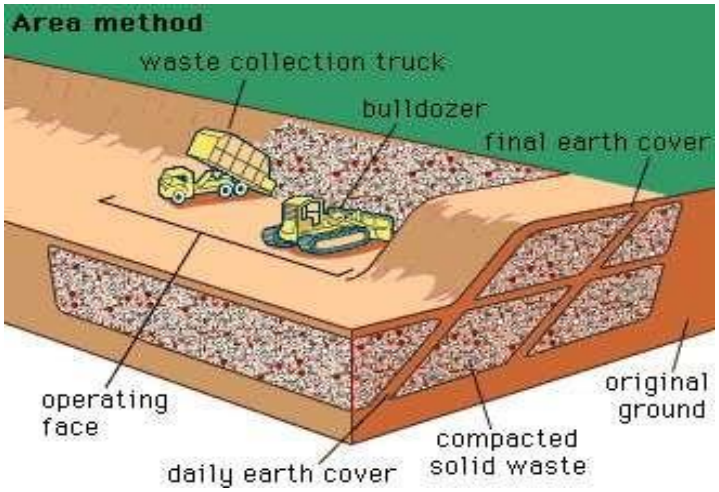
1. TRENCH METHOD

- The trench method consists of an excavated trench into which the solid wastes are spread, compacted and covered.
- The trench method is best suited for nearly level land where the water table is not near the surface.



2. AREA METHOD

- The area method is best suited for flat or gently sloping areas where some land depressions may exist.
- The wastes are spread, compacted and then covered.



3. RAMP METHOD

- It is also know as depression method.
- The slope or ramp is sometimes used in combination with the other two methods. The wastes are spread on an existing slope, compacted and covered.

➢ <u>Advantages :</u>

✓ Landfill site is a cheap waste disposal option.

✓ The gases given off by the landfill site could be collected and used for generating power.

✓ Lots of different types of waste can be disposed of by landfill in comparison to other waste disposal method.

➢ <u>Disadvantages :</u>

✓ Landfills can pollute air, water and also the soil.

✓ Landfill can contribute to the global warming.

ii. Landfills:

•A landfill may also refer to the ground that has been filled in with soil and rocks instead of waste materials, so that it can be used for a specific purpose, such as for building houses.

•Landfill, also known as a dump or tip, is a site for disposal of waste materials by burial.

•Older, poorly designed or poorly managed landfills can create a number of adverse environmental impacts such as wind-blown litter, attraction of vermin, and generation of liquid leachate.

[Ref: Indu Shekhar Thakur- Environmental Biotechnology, T. Srinivas, Environmental Biotechnology]

➢ Sanitary landfills:

•Sanitary Landfills are designed to greatly reduce or eliminate the risks that waste disposal may pose to the public health and environmental quality.

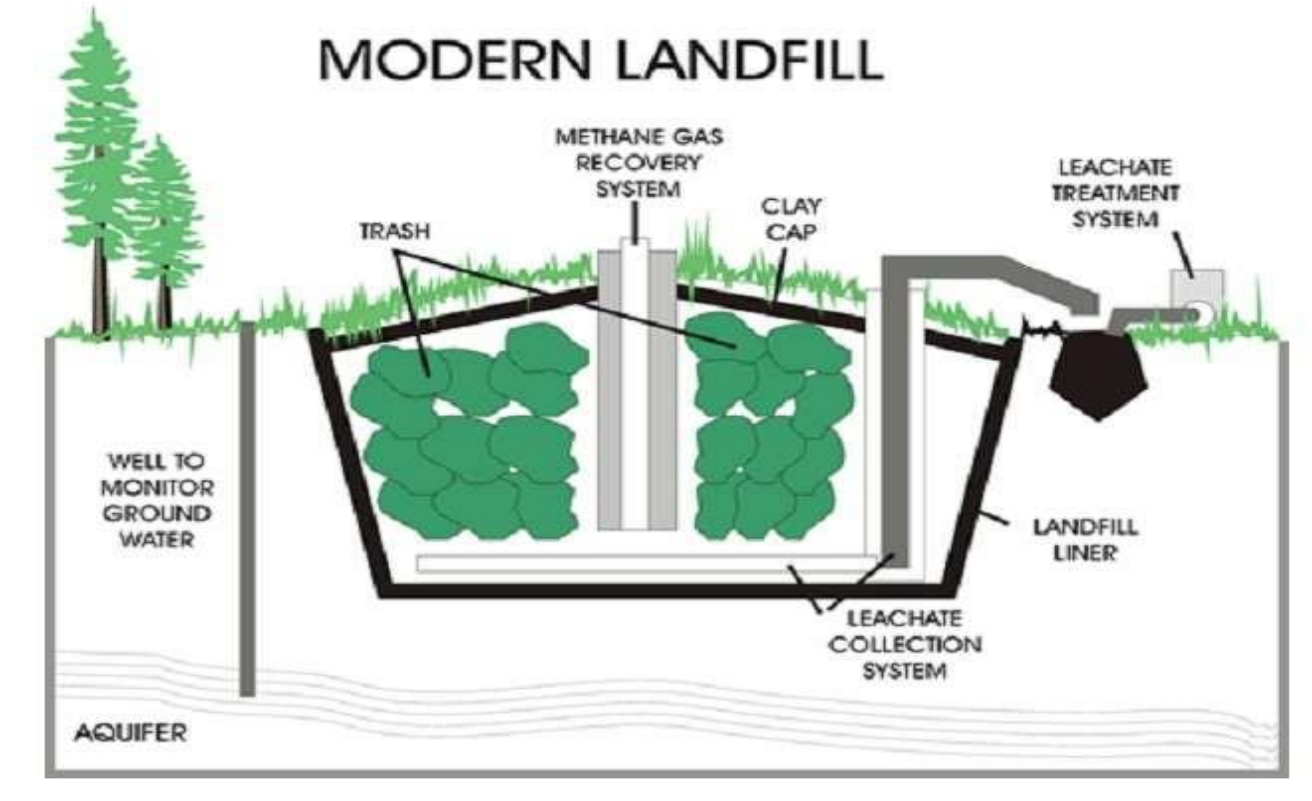


Fig: Main features of a modern landfill (Sanitary landfill)

[Ref: Indu Shekhar Thakur- Environmental Biotechnology and www.cyen.org]

1.LAND FILL

- It is the most traditional method of waste disposal.
- Waste is directly dumped into disused quarries, mining voids or borrow pits.
- Disposed waste is compacted and covered with soil
- Gases generated by the decomposing waste materials are often burnt to generate power.
- It is generally used for domestic waste.



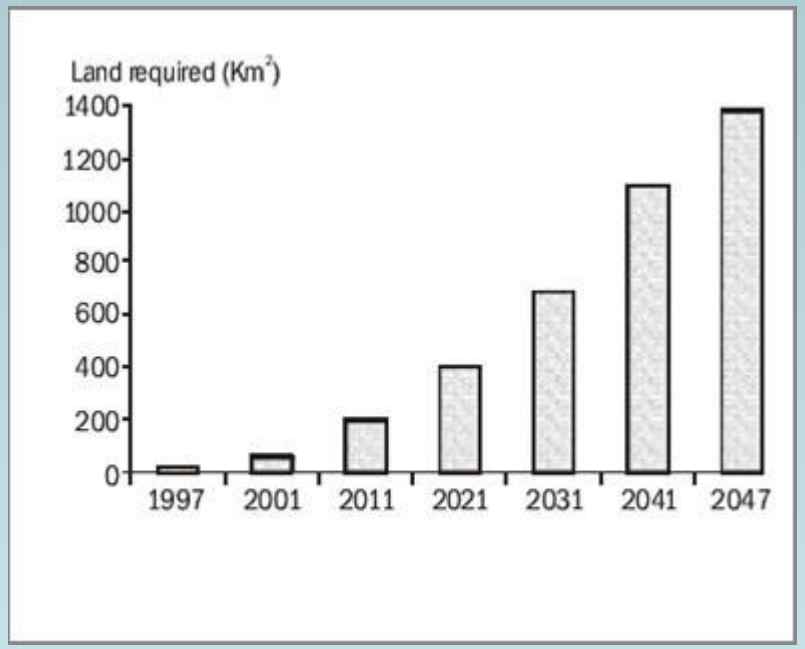
ADVANTAGES

- Landfill site is a cheap waste disposal option for the local council.
- Jobs will be created for local people.
- Lots of different types of waste can be disposed of by landfill in comparison to other waste disposal methods.
- The gases given off by the landfill site could be collected and used for generating power.

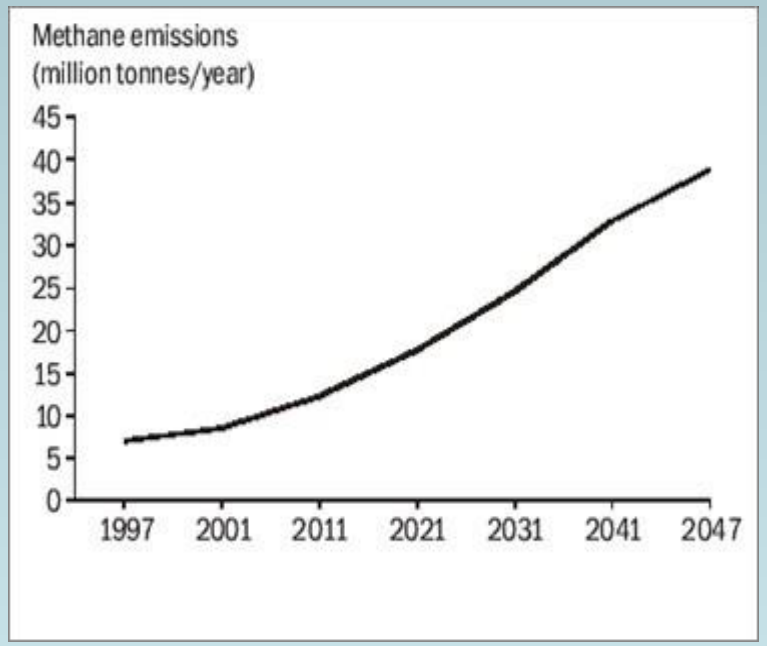
DISADVANTAGES

- The site will look ugly while it is being used for landfill.
- Dangerous gases are given off from landfill sites that cause local air pollution and contribute to global warming.
- Local streams could become polluted with toxins seeping through the ground from the landfill site.
- Once the site has been filled it might not be able to be used for redevelopment as it might be too polluted.





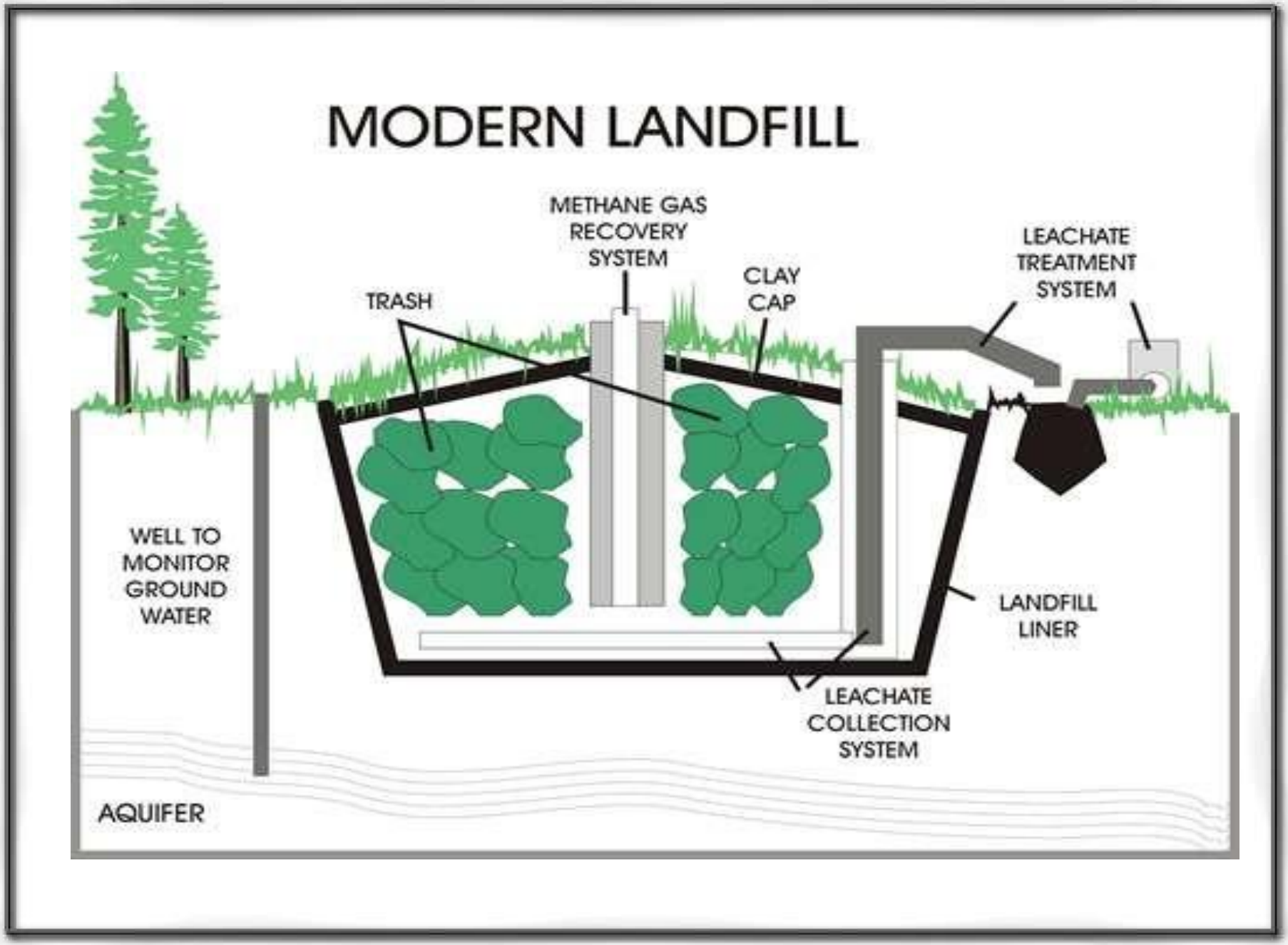
LAND REQUIRED FOR DISPOSAL OF MSW

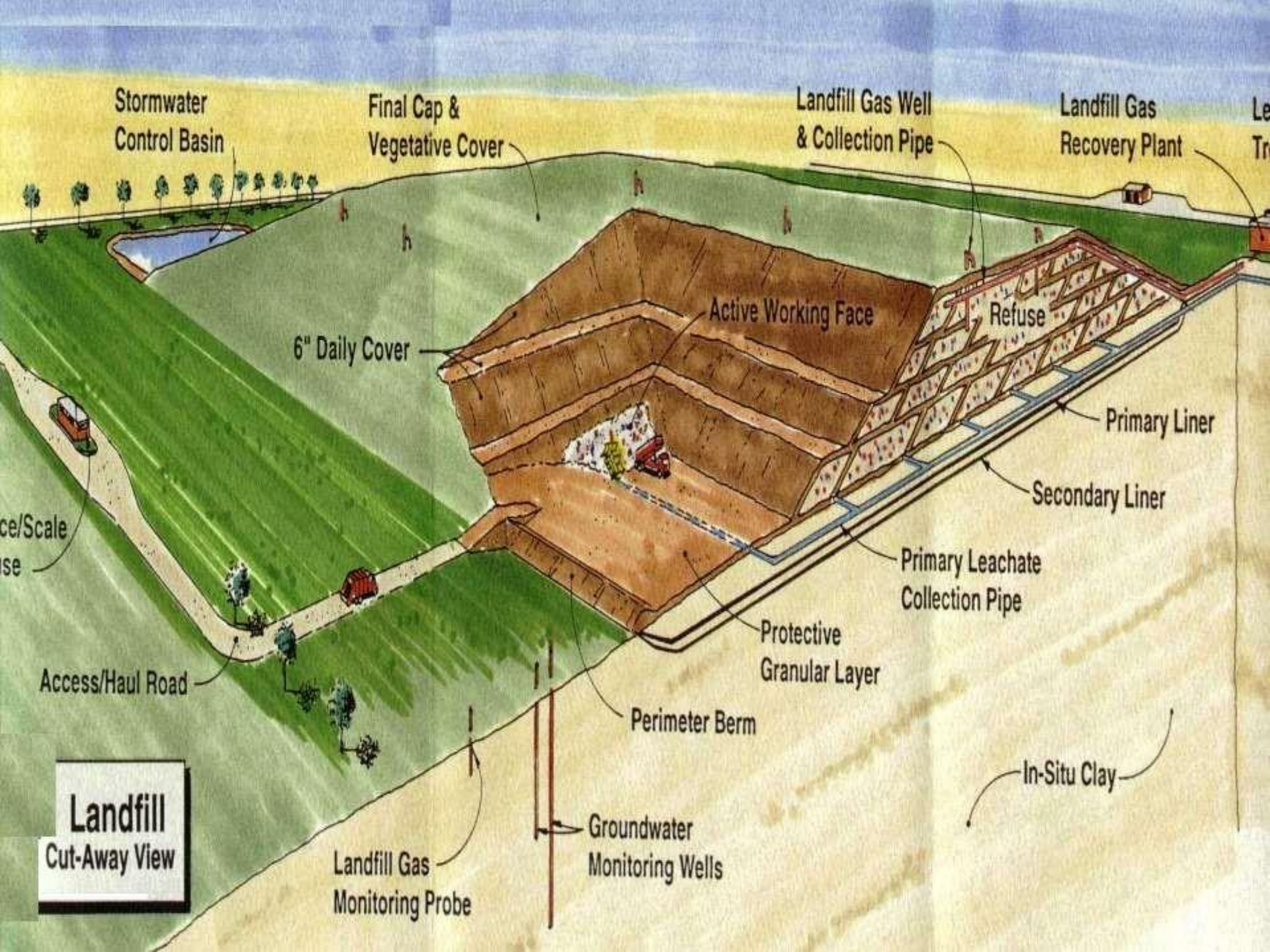


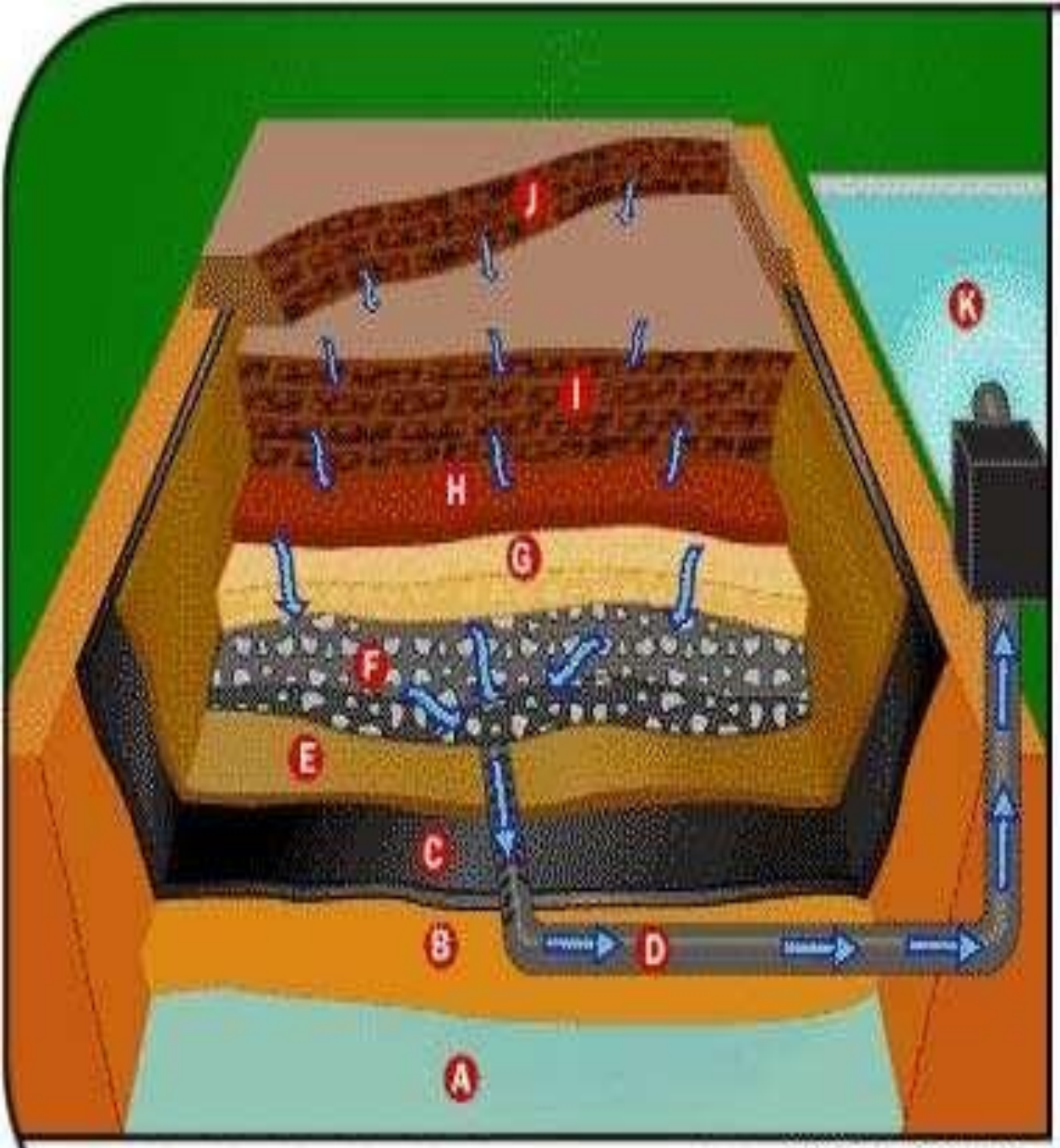
EMMISION OF METHANE FROM LANDFILL

Landfilling

- A landfill site is a site for the disposal of waste materials by burial and is the oldest form of waste treatment.
- Historically, landfills have been the most common methods of organized waste disposal and remain so in many places around the world.
- The dumping is done with layers of 1- 2 m.
- The layer is covered with soil of 20 cm thickness.


MODERN LANDFILL
METHANE GAS RECOVERY SYSTEM
TRASH
CLAY CAP
LEACHATE TREATMENT SYSTEM
WELL TO MONITOR GROUND WATER
LANDFILL LINER
LEACHATE COLLECTION SYSTEM
AQUIFER


Stormwater
Control Basin
Final Cap &
Vegetative Cover
Landfill Gas Well
& Collection Pipe
Landfill Gas
Recovery Plant
Active Working Face
Refuse
6" Daily Cover
Primary Liner
Secondary Liner
Primary Leachate
Collection Pipe
Protective
Granular Layer
Access/Haul Road
Perimeter Berm
In-Situ Clay
Groundwater
Monitoring Wells
Landfill Gas
Monitoring Probe
Landfill
Cut-Away View



A Ground Water
B Compacted Clay
C Plastic Liner
D Leachate Collection Pipe
E Geotextile Mat
F Gravel
G Drainage Layer
H Soil Layer
I Old Cells
J New Cells
K Leachate Pond

©2000 How Stuff Works

SINK HOLE

Advantages

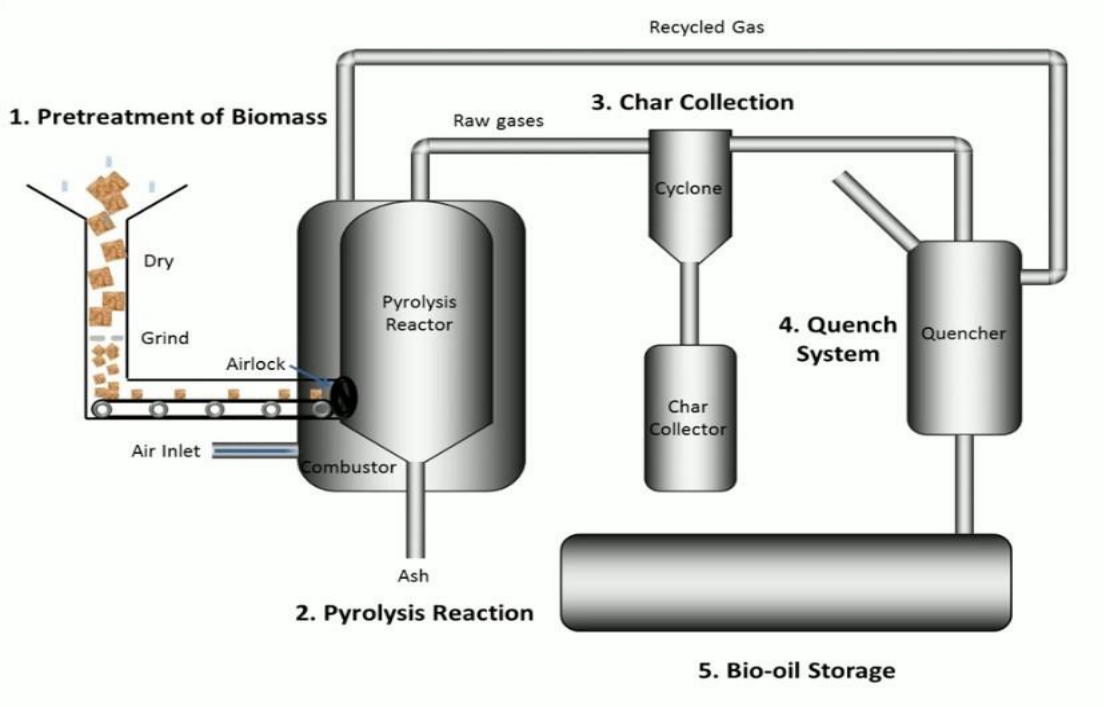
- Simple method.
- No costly plant required.
- No residues or by products need to be disposed.
- Separation not required.
- Unused land can be used.
- Methane gas can be used ass fuel.

Disadvantages

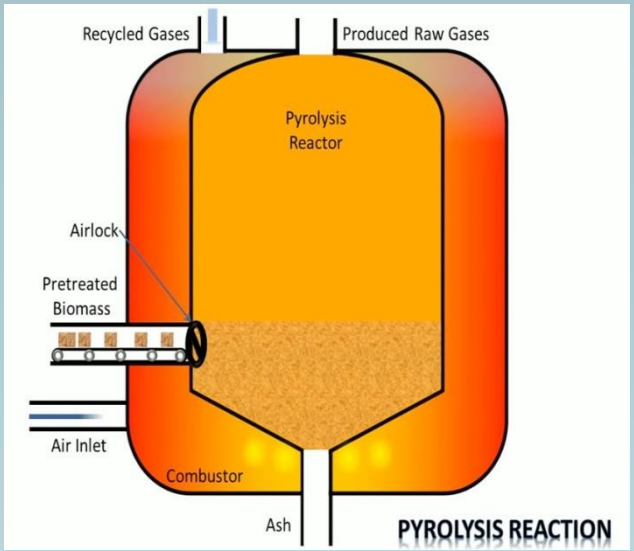
- Large land required.
- Proper dumping site may not be available.
- Odor problem.
- Use of insecticides required.
- Leachate should be collected regularly.
- Methane gas should be collected properly.
- Green house gas problem.

PYROLYSIS:-

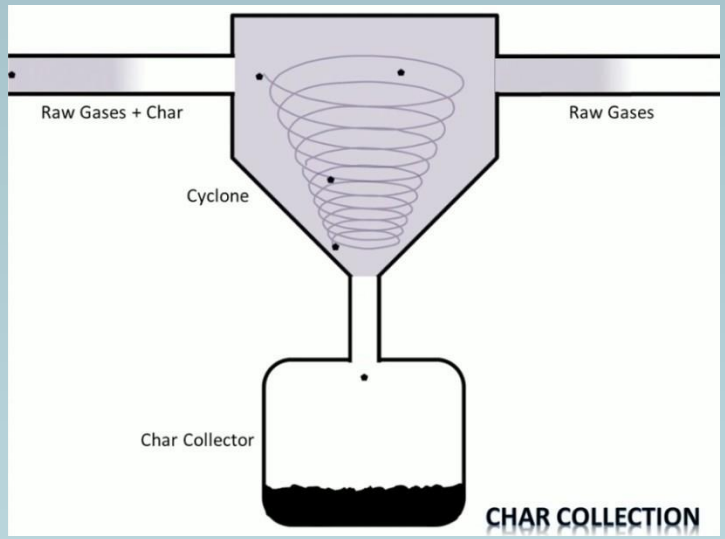
- Pyrolysis is defined as thermal degradation of waste in the absence of air to produce char, pyrolysis oil and syngas, e.g. the conversion of wood to charcoal also it is defined as destructive distillation of waste in the absence of oxygen. External source of heat is employed in this process.



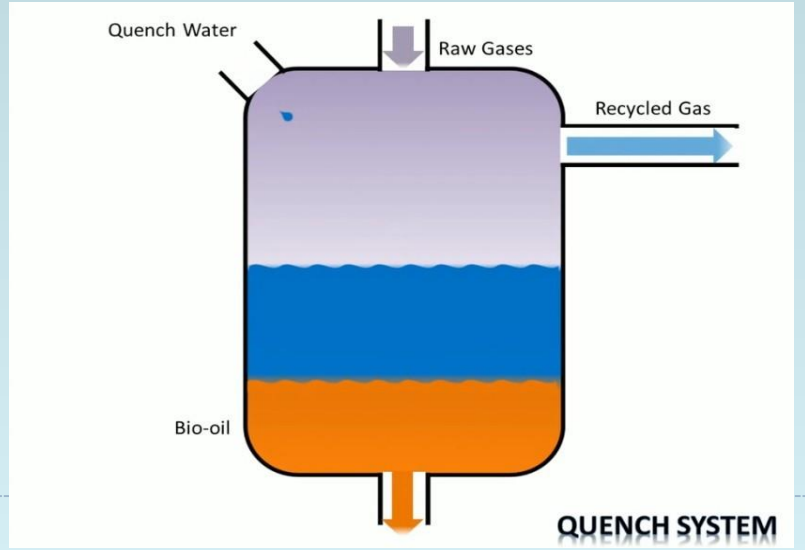
1


Recycled Gases
Produced Raw Gases
Pyrolysis Reactor
Airlock
Pretreated Biomass
Air Inlet
Combustor
Ash
PYROLYSIS REACTION

2

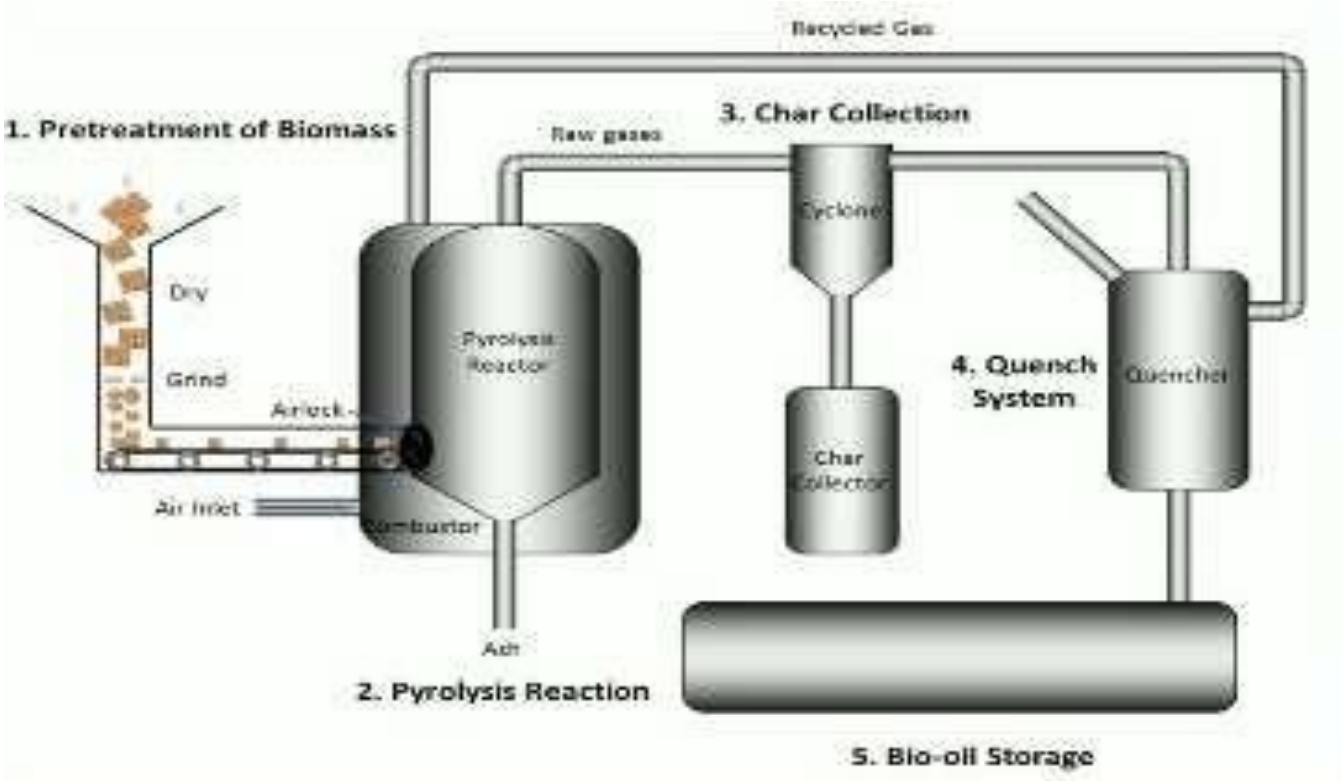

Raw Gases + Char
Raw Gases
Cyclone
Char Collector
CHAR COLLECTION

3


Quench Water
Raw Gases
Recycled Gas
Bio-oil
QUENCH SYSTEM

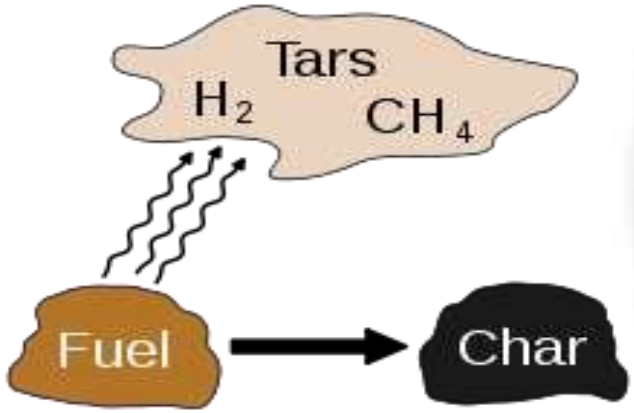
D. PYROLYSIS

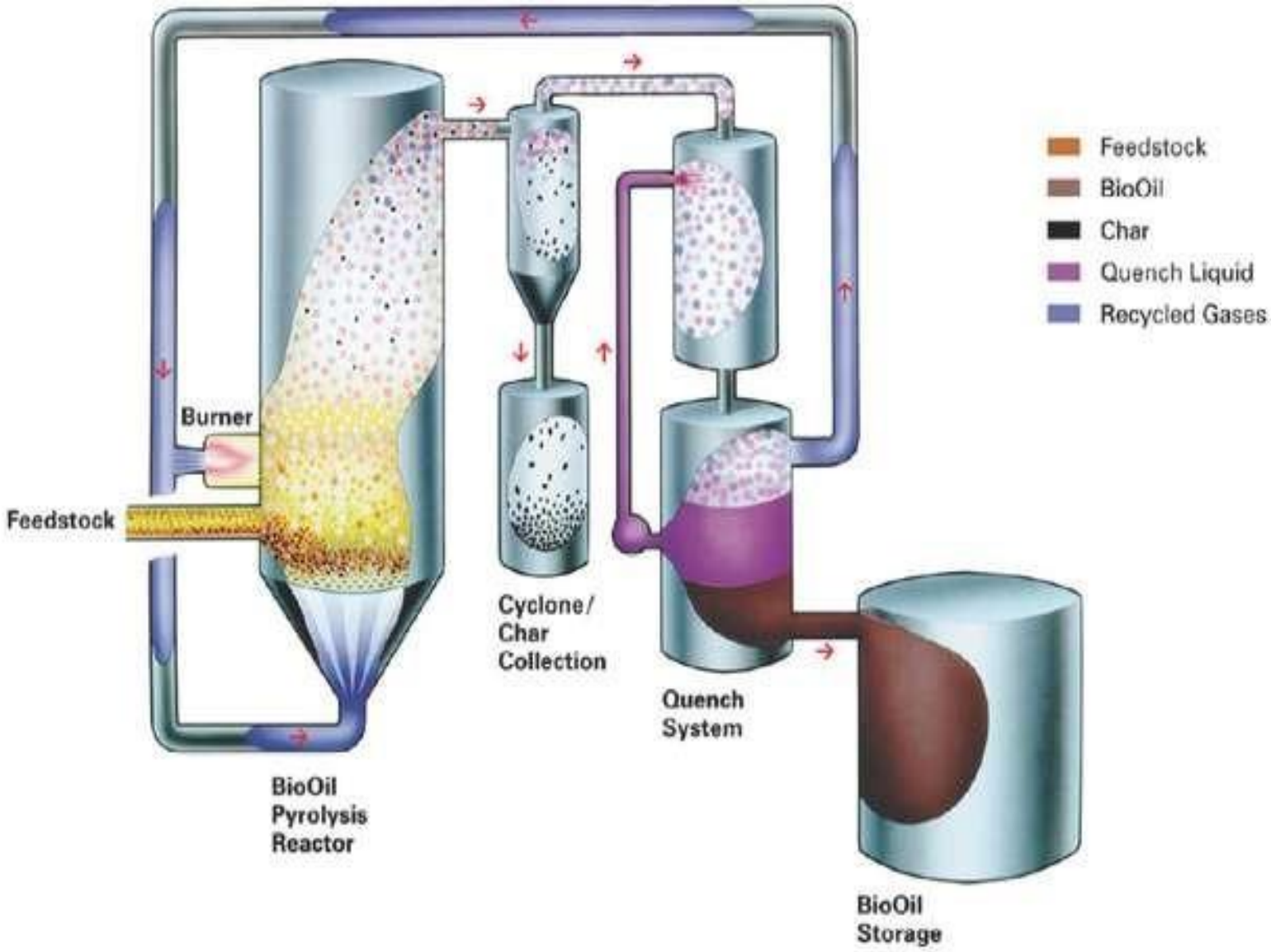
- Pyrolysis is a thermochemical decomposition of organic material at elevated temperatures in the absence of oxygen.
- It involves the simultaneous change of chemical composition and physical phase, and is irreversible.
- External heat source is employed.it yields gaseous, liquid and solid fractions.



Pyrolysis

- Heating of the solid waste at very high temp. in absence of air.
- Carried out at temp. between 500 °C – 1000 °C.
- Gas, liquid and chars are the by products.




Feedstock
BioOil
Char
Quench Liquid
Recycled Gases
Burner
Feedstock
Cyclone/
Char
Collection
Quench
System
BioOil
Pyrolysis
Reactor
BioOil
Storage

iv. Anaerobic Digestion:

•Anaerobic digestion is a regulated version of the natural events of landfill, in that it results in the controlled release of methane-rich biogas, which offers the potential for a very real form of energy from waste.

• It is carried out in large fermented tanks.

•In these tanks, solid waste is taken in the absence of oxygen and the anaerobic bacteria convert the large organic molecules mainly into methane CH_4 and carbon dioxide CO_2.

•Unlike composting, Anaerobic Digestion occurs at one of three distinct temperature ranges, namely:

- Cryophilic (<20 °C)
- Mesophilic (20–45 °C)
- Thermophilic (>45 °C)

[Ref: Gareth M. Evans, Environmental Biotechnology, Theory and Application]

➢ Micro-organisms involved in Anaerobic digestion:

There are four main groups of bacteria involved in Anaerobic digestion, as shown below, with some typical examples of each**:**

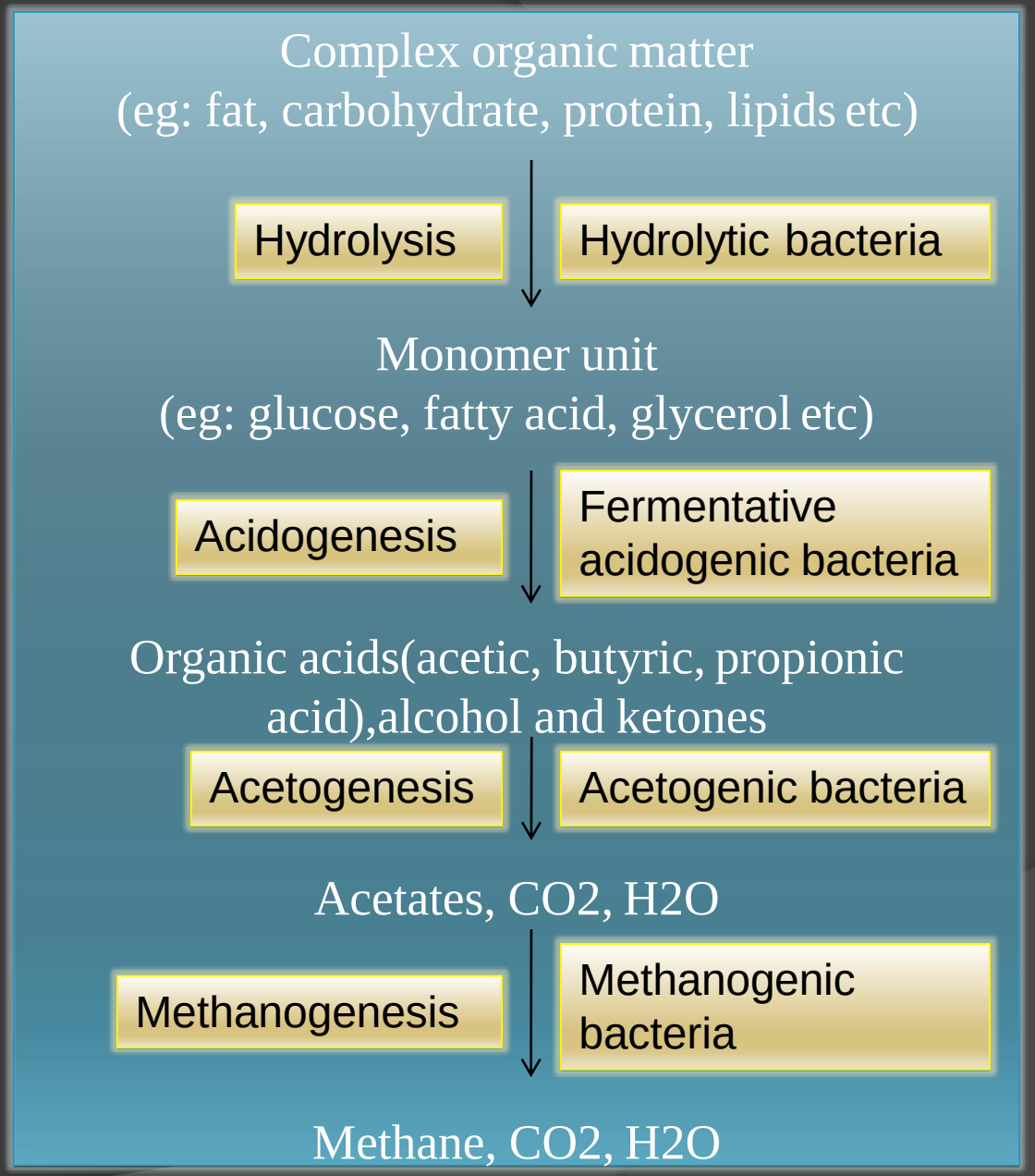
- **Hydrolytic fermentative bacteria** – *Clostridium and Peptococcus*.
- **Acetogenic bacteria** – *Syntrophobacter and Syntrophomonas.*
- **Acidogenic bacteria** – *Methanosarcina and Methanothrix.*
- **Methanogenic bacteria** – *Methanobacterium and Methanobrevibac*terium.

[Ref: Gareth M. Evans, Environmental Biotechnology, Theory and Application]

➢ The digestion process:

The digestion process involves 4 steps. They are:

- ❑ Hydrolysis
- ❑ Acidogenesis
- ❑ Acetogenesis
- ❑ Methanogenesis



[Ref: Gareth M. Evans, Environmental Biotechnology, Theory and Application]

❑ Encapsulation:

Solid particulate waste material is coated with a thermosetting resin which is compressed and cured to form a rigid core. The rigid core is coated with a flexible thermoplastic resin to provide a sealed encapsulated waste agglomerate which can withstand moderate compressive loads.

Encapsulation method are of the types:

- ❑ **Microencapsulation**
- ❑ **Microencapsulation**

[Ref**:** Indu Shekhar Thakur- Environmental Biotechnology, www.google.com/patents/US4234632]


WASTE RECYCLING
Low Cost Rugs
Carpet Waste,
Cutouts & Rejects
Cardboard &
Paper Waste
Recycled Paper
Extrusion & Backing
Offcuts Waste
Ash
Plastic Tables & Chairs
Waste Yarn & Fibres
Low Cost Housing
Carpet Underlay

The 3 R's



- **REDUCE**
  - You can help by *PRECYCLING*. 1/3 of all garbage is packaging.
  - Buy things that are in packages that can be recycled or are made of recycled materials.
  - When you buy something small, say no thanks to a bag.
- **REUSE**
  - Many things can be reused before you throw them out.
  - Use coffee cans and cottage cheese containers for storage
  - Use backs of paper or backs of used envelopes for jotting notes Put
  - leftovers in resalable containers instead of using wraps and foil
  - Use old clothes as rags for cleaning instead of paper towels
  - Have a garage sale or donate clothes, books or toys that you don't use anymore
- **RECYCLE**
  - Each year we use:
    - 25 billion plastic containers
    - 30 billion bottles & jars
    - 65 billion aluminum cans
    - 100 billion pounds of paper

4. Management of Solid waste:

The fundamental objective of waste processing is to reduce the amount of wastes through recycling and disposal of waste in a way not to impair environmental conservation.

Four R's should be followed for waste management:

- **Refuse**
- **Reuse**
- **Recycle**
- **Reduce**

[Ref: Indu Shekhar Thakur- Environmental Biotechnology]

The management of the solid waste is done according to its nature:

i. Management of Medical solid waste
ii. Management of non-degradable solid waste
iii. Management of Hazardous waste
iv. Management of non-hazardous & biodegradable solid waste
v. Management of electronic waste “e-waste”

[Ref: Indu Shekhar Thakur- Environmental Biotechnology]

i. Management of Medical solid waste:

•Hospital waste is generated during the diagnosis, treatment, or immunization of human beings or animals or in research activities in these fields or in the production or testing of biological.

•Medical solid waste includes both non-hazardous and hazardous waste constituents.

- ✓ The non-hazardous waste
- ✓ Hazardous wastes



Medical solid waste

Treatment options:

- ❑ **Incineration**
- ❑ **Sanitary landfill**

[Ref: Indu Shekhar Thakur- Environmental Biotechnology]

ii. Management of non-degradable solid waste:

- Examples of non-degradable solid wastes are:
  - Ferrous & non-ferrous metals: Eg: Iron, Steel and Aluminium etc.
  - Glass
  - Plastics
  - Textiles

- **Treatment options:**
  - Recycling
  - Sanitary landfill
  - Incineration

[Ref: Indu Shekhar Thakur- Environmental Biotechnology]

iii. Management of Hazardous waste:

Following process applied in hazardous waste treatment:

- **Physical separation**
- **Gravity separation**
- **Dissolved air floatation**
- **Solvent extraction**
- **Sorption on activated carbon**

Treatment options:

- Thermal treatment
  - Incineration
- Biological treatment



Hazardous waste

[Ref: Indu Shekhar Thakur- Environmental Biotechnology]

iv. Management of non-hazardous & biodegradable solid waste:

•Non-hazardous solid waste is total waste including municipal waste, industrial waste, agricultural waste and sewage sludge.

•Following methods are followed by management of non-hazardous and biodegradable solid wastes:

- Open Dumps
- Landfills
- Anaerobic Digestion
- Composting
- Vermicomposting

[Ref: Indu Shekhar Thakur- Environmental Biotechnology, T. Srinivas, Environmental Biotechnology]

vi. Management of Electronic waste, "E-Waste":

•Electronic waste is of concern largely due to the toxicity and carcinogenicity of some of the substances if processed improperly. Toxic substances in electronic waste include lead, mercury and cadmium. Carcinogenic substances in electronic waste may include polychlorinated biphenyls (PCBs).

•A typical computer may contain more than 6% lead by weight. Capacitors, transformers, PVC insulated wires of polychlorinated biphenyls.

[Ref**:** Indu Shekhar Thakur- Environmental Biotechnology]

➢ Sources, Constituents and effect of E-waste:

Source of e-wastes	Constituent	Health effects
Solder in printed circuit boards, glass panels and gaskets in computer monitors	Lead (PB)	• Damage to central and peripheral nervous systems, blood systems and kidney damage. • Affects brain development of children.
Chip resistors and semiconductors	Cadmium (CD)	• Toxic irreversible effects on human health. • Accumulates in kidney and liver.
Relays and switches, printed circuit boards	Mercury (Hg)	• Chronic damage to the brain. • Respiratory and skin disorders due to bioaccumulation in fishes.
Corrosion protection of untreated and galvanized steel plates, decorator or hardner for steel housings	Hexavalent chromium (Cr) VI	• Asthmatic bronchitis. • DNA damage.

➢ Treatment options:

E-waste consists of Diverse items like ferrous and non-ferrous metals, glass, plastics, electronic components and other items. The potential treatment options based on this composition are given below:

- ❑ Encapsulation
- ❑ Incineration
- ❑ Sanitary landfill

[Ref: T. Srinivas, Environmental Biotechnology]

Reuse of the electronic waste in the form of a stool:

Designer **Rodrigo Alonso** created **N+ew Seats** to address this -- stools that are formed of a whole lot of electronic junk. This was covered back in 2007.

The idea behind N+EW isn’t the creation of a recyclable object, but the way to immortalize and to give a last use to objects that their only destination is contamination."



E-Waste Furniture

[Ref: www.treehugger.com]

5. Factors affecting solid waste management:

There are certain factors that affect the management of solid waste: management. They are:

- **Per capita income and status**
- **Climate and percentage moisture**
- **Systematic growth of city**
- **Status of the municipality**
- **Resources available**

[Ref: T. Srinivas, Environmental Biotechnology]

WASTE MANAGEMENT POLICIES

- Environment protection act, 1986
- Hazardous waste rule 1989
- Bio-medical waste rule 1998
- Municipal solid waste rule 2000
- Waste management act 1996
- Solid waste policy in India 2006